LiDAR SLAM with Multiple Object Tracking(MOT) for Autonomous Driving

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Contents

目录

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Abstract

SLAM (Simultaneous Localization and Mapping) has become a fundamental technique for autonomous driving and robotic navigation due to its critical role in localization and environmental mapping [4]. Among LiDAR-based SLAM methods, FAST-LIO2 (Fast Direct LiDAR-inertial Odometry [3]) demonstrates excellent real-time performance by directly registering raw point clouds to the map and tightly coupling LiDAR and IMU (Inertial Measurement Unit) measurements. However, in dynamic scenes, moving objects such as vehicles and pedestrians introduce non-static geometric structures into the environment, which may degrade scan matching quality and lead to inaccurate trajectory estimation. Traditional approaches often treat dynamic objects as outliers and simply remove them, but such strategies may discard useful motion-related information and remain insufficient in complex traffic environments.

To address this issue, this paper proposes a trajectory optimization method for FAST-LIO2 in dynamic scenes via dynamic-object weight optimization and joint optimization. The proposed framework integrates object detection and MOT (Multi-Object Tracking) with the original FAST-LIO2 pipeline. In the front end, tracked object information is used to identify dynamic regions and assign adaptive weights to different point cloud observations, thereby reducing the negative influence of moving objects on pose estimation while preserving

摘要

同步定位与地图构建（SLAM）技术因其在定位和环境建图中的关键作用，已成为自动驾驶和机器人导航领域的基础技术[4]。在基于激光雷达的SLAM方法中，FAST-LIO2（快速直接激光雷达-惯性里程计[3]）通过直接将原始点云与地图配准，并紧密耦合激光雷达与惯性测量单元（IMU）的测量数据，展现出卓越的实时性能。然而在动态场景中，车辆、行人等移动物体会在环境中引入非静态几何结构，这可能导致扫描匹配质量下降并引发轨迹估算偏差。传统方法通常将动态物体视为异常值直接剔除，但此类策略容易丢失关键运动信息，在复杂交通环境中仍存在局限性。

为解决这一问题，本文提出了一种基于动态物体权重优化与联合优化的动态场景下FAST-LIO2轨迹优化方法。该框架将目标检测与多目标跟踪（MOT）技术与原始FAST-LIO2流程相结合。在前端处理阶段，通过利用被跟踪物体信息识别动态区域，并为不同点云观测数据分配自适应权重，从而在保持系统性能的同时有效降低移动物体对位姿估计的负面影响。

informative scene structure. In the back end, a joint optimization strategy is further introduced to refine ego trajectory estimation within a sliding window. Specifically, odometry consistency constraints and object observation consistency constraints are jointly constructed, enabling tracked object observations to provide additional guidance for trajectory correction.

Experiments on representative dynamic driving sequences demonstrate that the proposed method can effectively improve trajectory estimation accuracy compared with the baseline FAST-LIO2. The results further show that the method is especially beneficial in scenes with stronger dynamic interference, where front-end weight optimization and back-end joint optimization complement each other to enhance robustness and stability.

Keywords: SLAM; LiDAR-inertial odometry; FAST-LIO2; dynamic scenes; trajectory optimization; weight optimization; joint optimization

信息性场景结构。在后端系统中，进一步引入联合优化策略以在滑动窗口内优化自我轨迹估计。具体而言，通过联合构建里程计一致性约束与物体观测一致性约束，使被追踪物体的观测数据能够为轨迹修正提供额外指导。

针对典型动态驾驶序列的实验表明，与基准方法FAST-LIO2相比，所提出的方法能有效提升轨迹估计精度。研究结果进一步显示，该方法在动态干扰较强的场景中优势显著，其前端权重优化与后端联合优化机制相辅相成，显著增强了系统的鲁棒性和稳定性。

关键词：SLAM；激光雷达惯性里程计；FAST-LIO2；动态场景；轨迹优化；权重优化；联合优化

Chapter 1 Introduction

Section 1.1 Research Background

SLAM has become one of the fundamental technologies in autonomous driving and robotic navigation, as it enables a system to estimate its own motion while constructing a representation of the surrounding environment [12]. In practical intelligent driving systems, accurate localization is essential for perception, planning, and control. If the estimated trajectory is unreliable, errors may propagate to subsequent modules and reduce the overall stability of the system. Therefore, localization and mapping remain key research topics in autonomous systems and continue to attract broad attention in both academia and engineering applications.

Among different sensing modalities, LiDAR-based methods have shown significant advantages in outdoor localization and mapping tasks. Compared with cameras, LiDAR sensors directly provide three-dimensional geometric information of the environment and are less sensitive to illumination changes and visual disturbances. This makes them particularly suitable for autonomous driving scenarios, where vehicles need to operate in roads, residential areas, intersections, and other complex outdoor environments. Through continuous laser scanning, LiDAR can capture the spatial distribution of surrounding structures and generate point cloud data for pose estimation and map

第1章 引言

第1.1节 研究背景

SLAM技术已成为自动驾驶与机器人导航领域的核心技术之一，它使系统能够在构建周围环境表征的同时实现自主运动估计[12]。在实际智能驾驶系统中，精准的定位能力对感知、规划和控制环节至关重要。若估算出的运动轨迹存在偏差，误差可能向后续模块传递，导致系统整体稳定性下降。因此，定位与地图构建始终是自动驾驶系统的核心研究课题，在学术界和工程应用领域持续引发广泛关注。

在各类感知技术中，基于激光雷达（LiDAR）的方法在户外定位与地图构建任务中展现出显著优势。相较于传统摄像头，LiDAR传感器能直接获取环境的三维几何信息，且对光照变化和视觉干扰具有更强抗干扰能力。这使其特别适用于自动驾驶场景——车辆需要在道路、居民区、交叉路口等复杂户外环境中运行。通过持续激光扫描技术，LiDAR可精准捕捉周边建筑结构的空间分布特征，并生成用于姿态估计和地图构建的点云数据。

construction.

In many practical systems, LiDAR is further combined with inertial sensors to improve state estimation performance. The IMU provides high-frequency motion measurements, which help maintain motion continuity and compensate for the temporal limitations of LiDAR scans. By fusing LiDAR observations with inertial information, LiDAR-inertial methods can achieve better real-time performance and more stable pose estimation than LiDAR-only approaches. As a result, LiDAR-inertial localization and mapping has become an important research direction in autonomous driving and robotics.

Within this research direction, FAST-LIO2 is a representative method because of its efficiency and strong localization capability. Unlike conventional approaches that rely heavily on handcrafted feature extraction, FAST-LIO2 directly registers raw point clouds to the map and tightly couples LiDAR and inertial measurements for state estimation. This design preserves more geometric information from the original point cloud while also improving computational efficiency. In addition, its efficient map management strategy supports high-frequency processing and makes the framework suitable for real-time applications. Owing to these advantages, FAST-LIO2 has been widely regarded as an effective baseline for LiDAR-inertial odometry research.

However, real autonomous driving environments are often highly dynamic. Urban roads usually contain moving vehicles, pedestrians, cyclists, and other

建造

在众多实际应用系统中，激光雷达常与惯性传感器结合使用以提升状态估计性能。惯性测量单元（IMU）提供的高频运动数据有助于保持运动连续性，并弥补激光雷达扫描的时间分辨率局限。通过将激光雷达观测数据与惯性信息进行融合，激光雷达-惯性联合方法相比纯激光雷达方案能实现更优的实时性能和更稳定的姿态估计。因此，激光雷达-惯性联合定位与地图构建技术已成为自动驾驶与机器人领域的重要研究方向。

在该研究领域中，FAST-LIO2凭借其高效性与强大的定位能力成为代表性方法。与传统方法过度依赖人工特征提取不同，FAST-LIO2直接将原始点云与地图进行配准，并通过紧密耦合激光雷达与惯性测量数据实现状态估计。这种设计既保留了原始点云的更多几何信息，又提升了计算效率。此外，其高效的地图管理策略支持高频处理需求，使该框架适用于实时应用场景。基于这些优势，FAST-LIO2已被广泛视为激光雷达-惯性里程计研究领域的有效基准方法。

然而，真实的自动驾驶环境通常具有高度动态性。城市道路通常包含行驶中的车辆、行人、自行车骑行者及其他交通参与者。

traffic participants, while the spatial arrangement of the scene may change continuously due to traffic flow, temporary parking, or occlusion. These factors make real-world localization conditions much more complex than those in ideal static settings. Consequently, research on LiDAR SLAM in dynamic scenes has become increasingly important. For work built upon FAST-LIO2, improving trajectory estimation under such conditions is not only a meaningful extension of existing LiDAR-inertial research, but also of practical value for achieving more robust localization in realistic traffic environments.

Section 1.2 Problem Statement and Motivation

Although FAST-LIO2 has demonstrated strong performance in LiDAR-inertial odometry, its trajectory estimation accuracy may still degrade in dynamic scenes [5]. The core reason is that real traffic environments are not fully static. In practical autonomous driving scenarios, point clouds often contain observations from moving vehicles, pedestrians, cyclists, and other traffic participants. These dynamic elements continuously change their positions over time, which weakens the geometric consistency between consecutive observations and increases the difficulty of stable pose estimation.

For LiDAR-based localization, reliable trajectory estimation depends heavily on the assumption that most observed structures remain unchanged during motion. However, this assumption is often violated in urban

在交通场景中，参与者的位置信息会随交通流、临时停车或遮挡等因素持续变化，导致场景空间布局不断调整。这些因素使得现实世界中的定位条件比理想静态场景复杂得多。因此，针对动态场景中激光雷达SL-AM技术的研究变得愈发重要。基于FAST-LIO2框架开展的研究表明，在此类条件下提升轨迹估计精度不仅是对现有激光雷达惯性技术研究的合理延伸，更是实现真实交通环境中更稳健定位的关键实践价值。

第1.2节 问题陈述与动机

尽管FAST-LIO2在激光雷达惯性里程计领域展现出卓越性能，但在动态场景中其轨迹估计精度仍可能下降[5]。究其根本原因，真实交通环境并非完全静态。在实际自动驾驶场景中，点云数据通常包含移动车辆、行人、骑行者等交通参与者的观测数据。这些动态元素会持续改变位置坐标，导致连续观测数据间的几何一致性降低，从而增加了稳定位姿估计的难度。

对于基于激光雷达（LiDAR）的定位技术而言，可靠的轨迹估计高度依赖于一个假设：即大多数观测到的结构在运动过程中保持不变。然而，这一假设在城市环境中常被打破。

environments. When moving objects occupy a noticeable portion of the sensor's field of view, their point cloud observations may be mixed with static background structures during registration and map update [6]. As a result, the estimated motion of the ego vehicle can be affected by non-static geometric information, which may lead to local pose deviation and accumulated trajectory error. In scenes with dense traffic participants or frequent relative motion, this problem becomes more significant.

Another challenge lies in the complexity of dynamic scenes themselves. Dynamic interference is not limited to high-speed moving objects. Temporary stopping vehicles, slowly moving traffic participants, partial occlusions, and changes in local observation density may all influence the stability of point cloud registration [13]. In such cases, trajectory estimation errors may not come from a single obvious source, but from the combined effect of multiple dynamic factors. Therefore, improving localization performance in dynamic scenes requires more careful consideration of how scene changes affect geometric constraints in the odometry process.

From the perspective of practical autonomous driving, this issue is especially meaningful. Accurate ego trajectory estimation is a prerequisite for environmental perception, behavior planning, and motion control [1]. If localization becomes unstable in scenes with frequent dynamic interference, the reliability of the whole autonomous system may be reduced. Therefore, there is

在移动物体占据传感器视野显著区域时，其点云观测数据在配准与地图更新过程中可能与静态背景结构混杂[6]。这会导致自车运动估计受到非静态几何信息的影响，进而引发局部姿态偏差和累积轨迹误差。在存在密集交通参与者或频繁相对运动的场景中，该问题将更加显著。

另一个挑战在于动态场景本身的复杂性。动态干扰不仅限于高速移动物体，临时停车车辆、缓慢行驶的交通参与者、局部遮挡以及局部观测密度的变化都可能影响点云配准的稳定性[13]。在此类情况下，轨迹估计误差可能并非源自单一明显因素，而是多种动态因素共同作用的结果。因此，要在动态场景中提升定位性能，需要更细致地考量场景变化对里程计过程中几何约束条件的影响。

从自动驾驶的实际应用角度来看，这一问题具有特殊意义。精确的自我轨迹估计是环境感知、行为规划和运动控制的前提条件[1]。若在存在频繁动态干扰的场景中定位精度出现波动，整个自动驾驶系统的可靠性将显著降低。因此，

strong motivation to further study how FAST-LIO2 can be improved for dynamic environments, so that it can maintain better trajectory quality under more realistic road conditions.

Based on the above considerations, this work focuses on the trajectory estimation problem of FAST-LIO2 in dynamic scenes. The main motivation is to improve localization robustness and trajectory accuracy in the presence of dynamic interference, while preserving the efficiency and practical advantages of the original FAST-LIO2 framework. This problem setting is both theoretically meaningful for dynamic-scene LiDAR-inertial research and practically relevant for real-world autonomous driving applications.

Section 1.3 Related Development of Dynamic-Scene LiDAR SLAM

LiDAR SLAM and LiDAR-inertial odometry have been extensively studied for autonomous driving and robotic navigation. Early representative methods such as LOAM [2] established an effective framework for real-time lidar odometry and mapping by separating high-frequency odometry estimation from lower-frequency mapping. Building on this research line, later methods further improved computational efficiency, robustness, and estimation accuracy. Among them, FAST-LIO2 [3] has become a widely adopted baseline due to its direct registration of raw points to the map, tight coupling of LiDAR and inertial measurements, and efficient incremental map management. These

强烈希望进一步研究如何改进FAST-LIO2算法以适应动态环境，从而使其在更真实的道路条件下保持更高的轨迹质量。

基于上述考量，本研究聚焦于动态场景中FAST-LIO2的轨迹估计问题。核心目标是在动态干扰环境下提升定位鲁棒性与轨迹精度，同时保持原始FAST-LIO2框架的高效性与实际应用优势。该研究框架不仅对动态场景激光雷达-惯性联合研究具有理论意义，更在现实自动驾驶应用场景中展现出重要应用价值。

第1.3节 动态场景LiDAR SLAM的相关发展

激光雷达SLAM与激光雷达惯性里程计技术在自动驾驶和机器人导航领域已得到广泛研究。早期代表性方法如LOAM[2]通过将高频里程估计与低频地图构建分离，为实时激光雷达里程计与地图构建建立了有效框架。基于这一研究方向，后续方法在计算效率、鲁棒性及估计精度方面实现了进一步提升。其中，FAST-LIO2[3]凭借原始点与地图的直接配准、激光雷达与惯性测量的紧密耦合以及高效的增量地图管理机制，已成为被广泛采用的基准方法。

methods have demonstrated strong performance in structured environments and provided an important foundation for subsequent dynamic-scene research.

However, traditional LiDAR SLAM methods are mainly designed under the assumption that most observed structures are static. When deployed in urban traffic environments, moving vehicles, pedestrians, and other dynamic objects may introduce unstable geometric observations into the localization process. To address this issue, one major research direction has focused on identifying and removing dynamic observations from point clouds or maps. Representative methods such as Removert [7] and ERASOR [8] aim to construct static point cloud maps in dynamic urban environments by detecting dynamic regions and filtering out moving-object traces. These studies show that dynamic-object suppression can improve the consistency of static map construction and reduce the negative impact of moving objects on subsequent localization.

With the increasing demand for robust localization in realistic traffic scenarios, more recent studies have extended this idea from map-oriented dynamic removal to dynamic-scene LiDAR-inertial odometry. For example, some works explicitly incorporate moving-object detection into the odometry pipeline to improve pose estimation in dynamic scenes. A representative example is the work by Hu et al. [9] , which combines LiDAR-inertial odometry with moving-object detection for dynamic environments. This type of research indicates that dynamic-scene localization is not only a map-cleaning

这些方法在结构化环境中展现出优异性能，并为后续动态场景研究奠定了重要基础。

然而，传统激光雷达SLAM方法的设计主要基于“多数观测结构为静态”的假设。在城市交通环境中应用时，移动车辆、行人等动态物体可能为定位过程引入不稳定几何观测数据。为解决这一问题，研究重点聚焦于从点云或地图中识别并剔除动态观测数据。代表性方法如Removert [7]和 ERASOR[8]通过检测动态区域并过滤移动物体轨迹，致力于在动态城市环境中构建静态点云地图。研究表明，动态物体抑制技术不仅能提升静态地图构建的一致性，还可有效降低移动物体对后续定位过程的负面影响。

随着对真实交通场景中鲁棒定位需求的持续增长，近期研究将这一理念从地图导向的动态目标剔除扩展至动态场景激光雷达-惯性里程计技术。例如，部分研究将移动目标检测技术直接整合到里程计计算流程中，以提升动态场景中的位姿估计精度。代表性案例是胡等人[9]的研究成果，该研究将激光雷达-惯性里程计与移动目标检测技术相结合应用于动态环境场景。这类研究结果表明，动态场景定位技术不仅需要地图清理技术，更需要融合多源数据处理能力。

problem, but also a trajectory estimation problem in which dynamic observations directly influence odometry quality.

Another related development is the tighter integration of localization and object-level dynamic analysis. DL-SLOT [10] , for example, formulates dynamic lidar SLAM and object tracking within a unified optimization framework, showing that ego-motion estimation and dynamic-object tracking can be treated as closely related problems in road scenarios. This line of work suggests that, beyond suppressing dynamic interference, structured information associated with moving objects may also play a role in improving localization robustness.

Overall, existing research shows a clear development path in dynamic-scene LiDAR SLAM: from classical geometric odometry and mapping, to dynamic point removal and static map recovery, and then to methods that address dynamic-scene trajectory estimation more directly. Recent review work has also summarized this trend and highlighted the growing importance of dynamic-object filtering in LiDAR-based SLAM systems [11] . This development provides important background for further improving FAST-LIO2 in dynamic scenes and forms the research basis of the present work.

这不仅是一个问题，也是一个轨迹估计问题，其中动态观测数据会直接影响里程计测量质量。

另一项相关进展是定位技术与目标级动态分析的深度融合。以DL-SLOT[10]为例，该研究在统一优化框架下构建了动态激光雷达SLAM与目标跟踪系统，证明在道路场景中，自我运动估计与动态目标跟踪可被视为高度关联的优化问题。这一研究方向表明，除了抑制动态干扰外，移动物体所携带的结构化信息也可能对提升定位系统的鲁棒性起到关键作用。

总体而言，现有研究表明动态场景激光雷达SLAM技术的发展路径清晰可循：从经典的几何里程计与地图构建，到动态点位剔除与静态地图重建，再到更直接针对动态场景轨迹估计的方法。近期综述研究也总结了这一发展趋势，并强调了动态目标滤波在激光雷达SLAM系统中的重要性日益凸显[11]。这一技术演进为优化FAST-LIO2在动态场景中的表现提供了重要理论基础，同时也构成了本研究的核心研究依据。

Section 1.4 Main Idea and Contributions of This Work

This work focuses on trajectory optimization for FAST-LIO2 in dynamic scenes. The proposed framework combines object detection, multi-object tracking, odometry estimation, and trajectory refinement into a unified pipeline for autonomous driving scenarios. Its main purpose is to improve ego trajectory estimation under dynamic interference while preserving the efficiency of the original FAST-LIO2 framework.

The workflow starts from front-end perception. For each frame, the system first performs 3D object detection on the point cloud and then uses multi-object tracking to maintain temporal consistency of the detected vehicles. At the same time, FAST-LIO2 processes LiDAR and IMU data to estimate the ego pose. In this way, the framework obtains both object-level scene information and real-time odometry information from the same driving process.

Based on these outputs, the framework further connects tracking and localization for trajectory-oriented optimization. The tracked objects provide structured dynamic-scene information for FAST-LIO2, which is used for front-end weight optimization to reduce the influence of unstable regions during odometry estimation. On this basis, a back-end joint optimization stage is further introduced to refine ego trajectory estimation and improve overall trajectory consistency in dynamic scenes.

Overall, this work builds a complete dynamic-scene trajectory optimization

第1.4节 本研究的主要观点与贡献

本研究聚焦于动态场景中FAST-LIO2的轨迹优化问题。所提出的框架将目标检测、多目标跟踪、里程计估计及轨迹优化等技术整合为统一处理流程，适用于自动驾驶场景。其核心目标是在保持原始FAST-LIO2框架效率的前提下，提升动态干扰环境下的自我轨迹估计精度。

该工作流程始于前端感知阶段。系统首先对每帧图像的点云数据进行三维目标检测，随后通过多目标跟踪技术确保检测车辆的时间一致性。与此同时，FAST-LIO2系统同步处理激光雷达（LiDAR）与惯性测量单元（IMU）数据以估算车辆本体姿态。通过这种机制，该框架能够在同一驾驶过程中同步获取目标级场景信息与实时里程计数据。

基于这些输出结果，该框架进一步将轨迹跟踪与定位技术相结合，以实现以轨迹为导向的优化。被跟踪对象为FAST-LIO2系统提供了结构化的动态场景信息，这些信息用于前端权重优化，从而降低里程计估计过程中不稳定区域的影响。在此基础上，系统引入了后端联合优化阶段，以精炼自我轨迹估计精度，并提升动态场景中的整体轨迹一致性。

总体而言，本研究构建了一个完整的动态场景轨迹优化模型

pipeline for FAST-LIO2. By connecting detection, tracking, front-end weight optimization, and back-end joint optimization within one framework, the proposed study extends conventional LiDAR-inertial trajectory estimation to a more structured dynamic-scene setting and provides a practical solution for improving localization robustness in realistic traffic environments.

FAST-LIO2系统数据处理流程。本研究通过将检测、跟踪、前端权重优化与后端联合优化整合至统一框架，将传统激光雷达-惯性轨迹估计方法拓展至更具结构化的动态场景环境，为提升真实交通场景中的定位鲁棒性提供了实用解决方案。

Chapter 2 Related Theories and Key Techniques

Section 2.1 Overview of FAST-LIO2

FAST-LIO2 [3] is a representative LiDAR-inertial odometry framework designed for real-time and accurate state estimation in complex environments. By tightly coupling LiDAR measurements with inertial information, it achieves high-frequency pose estimation while maintaining strong robustness in large-scale outdoor scenes. Owing to its efficiency and reliable localization capability, FAST-LIO2 has become an important baseline for many studies related to LiDAR-inertial localization and mapping.

As illustrated in the system framework in Figure 2-1, FAST-LIO2 mainly consists of two closely connected parts: state estimation and map management. The state estimation module fuses LiDAR and inertial measurements to estimate the motion state of the platform, while the mapping module incrementally updates the local map to support point cloud registration. Through this tightly coupled design, the framework maintains a balance between estimation accuracy and real-time performance.

第二章 相关理论与关键技术

第2.1节 FAST-LIO2概述

FAST-LIO2[3]是专为复杂环境实时精准状态估计而设计的代表性激光雷达-惯性里程计框架。通过将激光雷达测量数据与惯性信息进行深度耦合，该框架在保持大规模户外场景中强鲁棒性的同时，实现了高频位姿估计。凭借其高效定位能力和可靠定位性能，FAST-LIO2已成为激光雷达-惯性定位与地图构建领域多项研究的重要基准模型。

如图2-1所示系统架构，FAST-LIO2主要由两个紧密关联的模块构成：状态估计模块与地图管理模块。状态估计模块通过融合激光雷达数据与惯性测量数据来估算平台运动状态，而地图管理模块则通过增量式更新局部地图以支持点云配准。这种紧密耦合的设计架构在保持估计精度的同时，实现了实时性能与系统响应之间的动态平衡。

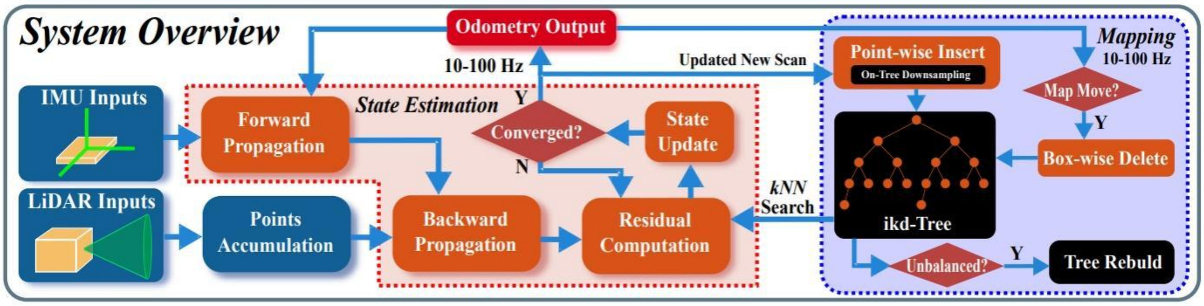

Figure 2-1 FAST-LIO2 System Architecture

The core idea of FAST-LIO2 is to directly register raw point clouds to the map instead of relying heavily on handcrafted feature extraction. Compared with conventional LiDAR odometry methods that first extract edge or planar features and then perform matching, FAST-LIO2 makes fuller use of the geometric information contained in the original point cloud. This design reduces preprocessing complexity and improves computational efficiency, which is particularly beneficial for applications requiring real-time performance. As shown in Figure 2-2, the incoming point cloud is directly aligned with the map, allowing the framework to avoid excessive dependence on manually selected geometric features.

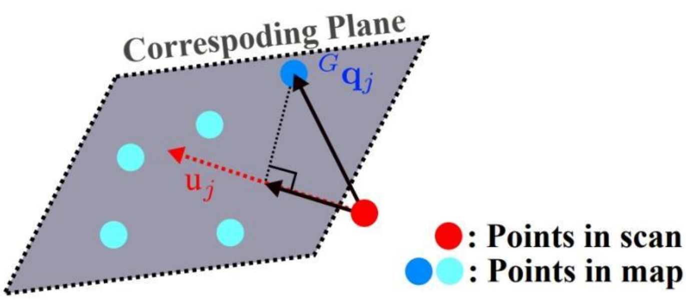

Figure 2-2 Direct Point Cloud Registration

In the FAST-LIO2 framework, the IMU provides high-frequency motion measurements for state propagation, while the LiDAR supplies geometric

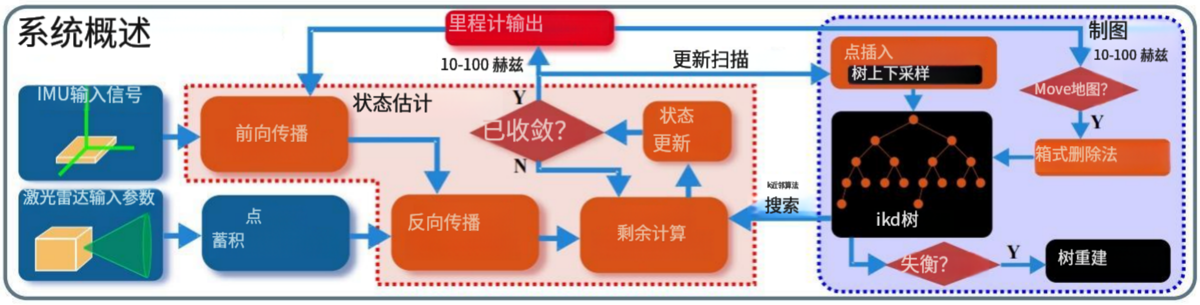

图2-1 FAST-LIO2系统架构

FAST-LIO2的核心理念在于直接将原始点云与地图进行配准，而非过度依赖人工特征提取。相较于传统激光雷达里程计方法需要先提取边缘或平面特征再进行匹配，FAST-LIO2能更充分地利用原始点云中蕴含的几何信息。这种设计有效降低了预处理复杂度并提升了计算效率，对于需要实时性能的应用场景具有显著优势。如图2-2所示，输入点云可直接与地图进行对齐，使得该框架能够减少对人工选取几何特征的过度依赖。

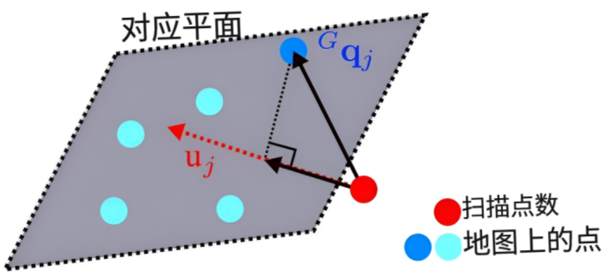

图2-2 直接点云配准

在FAST-LIO2框架中，惯性测量单元（IMU）为状态传播提供高频运动测量数据，而激光雷达（LiDAR）则提供几何信息。

observations of the surrounding environment. The inertial measurements are first used to predict the motion state of the platform, providing an initial estimate for the current scan. Then, the incoming point cloud is registered to the map, and the residuals between the observed points and the local map are used to update the system state. Through this tightly coupled process, FAST-LIO2 combines the short-term motion continuity of inertial sensing with the structural constraints of LiDAR observations, enabling accurate and stable ego-motion estimation.

Another important feature of FAST-LIO2 is its efficient map management mechanism based on the incremental k-d tree. As new LiDAR scans arrive, map points can be inserted and updated incrementally, which allows the system to maintain a local map for fast nearest-neighbor search and point-to-map registration. This strategy significantly improves runtime efficiency compared with repeatedly rebuilding the map structure from scratch. As illustrated in Figure 2-3, the incremental update strategy of the ikd-Tree enables efficient point insertion, deletion, and reorganization during mapping, thereby supporting high-frequency localization and map maintenance.

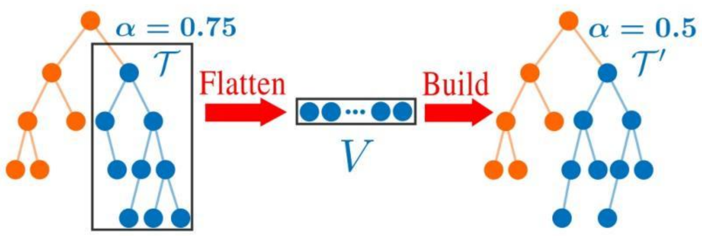

Figure 2-3 Optimization Strategy of ikd-Tree

通过环境观测数据，惯性测量系统首先用于预测平台运动状态，为当前扫描任务提供初始估计值。随后将传入的点云数据与地图进行配准，利用观测点与局部地图之间的残差信息更新系统状态。通过这种紧密耦合的处理流程，FAST-LIO2系统将惯性传感技术的短期运动连续性与激光雷达观测的结构约束条件相结合，实现了精准稳定的本体运动估计。

FAST-LIO2的另一大亮点在于其基于增量k-d树的高效地图管理机制。当新的激光雷达扫描数据传入时，系统能够通过增量方式插入并更新地图点位，从而实时维护局部地图数据，实现快速邻近点搜索和点位与地图的精准配准。相较于需要反复从头构建地图结构的传统方法，这种策略显著提升了运行效率。如图2-3所示，ikd树的增量更新机制在测绘过程中实现了点位的高效插入、删除与重组操作，有效支持高频次定位需求和地图维护工作。

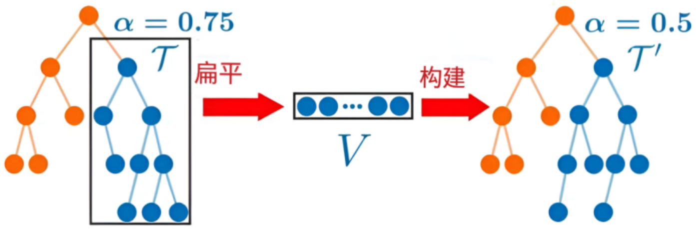

图2-3 ikd-Tree优化策略

From the perspective of system performance, FAST-LIO2 offers several important advantages. First, its direct registration strategy avoids excessive dependence on handcrafted feature extraction and makes the framework more efficient in dense outdoor environments. Second, the tight coupling between LiDAR and inertial data enhances motion estimation stability, especially during rapid movement. Third, the incremental map update strategy further improves real-time capability and makes the framework suitable for practical autonomous driving and robotic applications.

At the same time, FAST-LIO2 is still essentially developed under the assumption that most observed structures are sufficiently stable for reliable registration. In highly dynamic environments, moving vehicles and other traffic participants may introduce non-static observations into the localization process, which can influence map consistency and degrade trajectory estimation quality. For this reason, although FAST-LIO2 provides a strong and efficient baseline, further trajectory-oriented optimization is still meaningful when the framework is deployed in dynamic driving scenes.

Overall, FAST-LIO2 provides the fundamental odometry and mapping basis for the present work. Its efficient LiDAR-inertial coupling, direct point-to-map registration, and incremental map management make it well suited as the core localization framework upon which further dynamic-scene trajectory optimization can be built.

从系统性能角度来看，FAST-LIO2展现出多项显著优势。首先，其直接配准策略有效避免了对人工特征提取的过度依赖，使该框架在高密度户外环境中运行效率显著提升。其次，激光雷达与惯性数据间的紧密耦合增强了运动估计稳定性，尤其在快速移动场景中表现突出。第三，增量式地图更新策略进一步优化了实时处理能力，使该框架完美适配自动驾驶与机器人技术的实际应用场景。

与此同时，FAST-LIO2的核心开发理念仍基于一个基本假设：大多数观测到的结构具有足够的稳定性，能够实现可靠的配准。在高度动态的环境中，移动车辆及其他交通参与者可能为定位过程引入非静态观测数据，这不仅会影响地图一致性，还会降低轨迹估计质量。因此，尽管FAST-LIO2提供了强大且高效的基准框架，但在动态驾驶场景中部署该系统时，进行以轨迹优化为导向的进一步改进仍然具有重要价值。

总体而言，FAST-LIO2为本研究提供了基础里程计测量与地图构建的基础。其高效的激光雷达-惯性耦合技术、直接点对地图配准以及增量式地图管理机制，使其成为构建动态场景轨迹优化所需的核心定位框架的理想选择。

Section 2.2 Principles of 3D Object Detection

3D object detection is an important component in autonomous driving perception, since it provides structured information about surrounding traffic participants such as vehicles, pedestrians, and cyclists. Compared with raw point cloud observations, detection results offer a more compact and interpretable representation of the scene, including object category, position, size, and orientation. Such information is essential for subsequent tasks such as object tracking, scene understanding, and dynamic-scene analysis.

In LiDAR-based autonomous driving systems, 3D object detection aims to identify target objects directly from point cloud data and estimate their bounding boxes in three-dimensional space. Unlike image-based detection, LiDAR detection relies on geometric measurements rather than texture or color information, which makes it more robust to illumination changes and more suitable for outdoor environments [16] . However, point clouds are sparse, irregular, and non-uniformly distributed, which makes 3D detection more challenging than conventional 2D image detection.

To address these issues, many modern LiDAR detection methods first transform raw point clouds into a more regular representation before feature extraction. In this work, the detection module is built upon PointPillars [14] , which is a widely used real-time 3D object detection framework for point cloud data. The main idea of PointPillars is to divide the point cloud into vertical

第2.2节 三维目标检测原理

三维物体检测是自动驾驶感知系统中的关键组件，因其能提供关于周边交通参与者（如车辆、行人及骑行者）的结构化信息。相较于原始点云观测数据，检测结果能更紧凑且直观地呈现场景特征，包括物体类别、位置、尺寸及朝向等信息。这些数据对后续任务（如物体追踪、场景理解及动态场景分析）具有不可或缺的作用。

在基于激光雷达的自动驾驶系统中，三维目标检测旨在直接从点云数据中识别目标物体，并估算其在三维空间中的边界框。与基于图像的检测方法不同，激光雷达检测依赖几何测量而非纹理或颜色信息，这使其对光照变化更具鲁棒性，也更适合户外环境[16]。然而，点云数据具有稀疏性、不规则性和非均匀分布特征，这使得三维检测比传统二维图像检测更具挑战性。

为解决这些问题，许多现代激光雷达检测方法在进行特征提取前，会先将原始点云数据转换为更规则的表示形式。本研究中的检测模块基于PointPillars[14]构建，该框架是点云数据实时三维物体检测领域的通用解决方案。PointPillars的核心思想在于将点云数据划分为垂直方向的

columns, referred to as pillars, in the bird's-eye-view plane. Points within each pillar are encoded into a learned feature representation, allowing irregular point cloud data to be converted into a pseudo-image structure that can be efficiently processed by two-dimensional convolutional neural networks.

After pillar encoding, spatial features are extracted through a backbone network to generate high-level representations of the scene. Based on these features, the detection head predicts the category and 3D bounding box parameters of target objects. The output typically includes the class label, object location, box dimensions, and orientation. Through this process, PointPillars achieves a balance between computational efficiency and detection accuracy, making it suitable for autonomous driving applications that require real-time perception.

For the present work, 3D object detection serves as the first step in obtaining object-level scene information from dynamic driving environments. The detected vehicles provide the basis for subsequent multi-object tracking, enabling the system to establish temporal consistency of surrounding targets across consecutive frames. In this sense, the detection module is not only a perception component, but also an important foundation for later trajectory-oriented processing in dynamic scenes.

Overall, 3D object detection transforms unstructured point cloud observations into structured object representations and provides essential input

在鸟瞰视角平面上，这些柱状结构被称为支柱。各支柱内的点被编码为学习特征表示，从而将不规则点云数据转化为伪图像结构，该结构可被二维卷积神经网络高效处理。

柱状编码完成后，通过主干网络提取空间特征以生成场景的高层次表征。基于这些特征，检测头会预测目标物体的类别及三维边界框参数。输出结果通常包含类别标签、物体位置、边界框尺寸和方向信息。通过这一过程，PointPillars在计算效率与检测精度之间实现了平衡，使其特别适用于需要实时感知的自动驾驶应用场景。

在本研究中，三维物体检测是动态驾驶环境中获取物体级场景信息的第一步。检测到的车辆为后续多目标跟踪提供基础数据，使系统能够确保连续帧中周围目标的时间一致性。由此可见，检测模块不仅是感知组件，更是动态场景中后续轨迹导向处理的重要基础。

总体而言，三维目标检测技术将非结构化点云观测数据转化为结构化目标表征，并提供关键输入数据。

for downstream tracking and dynamic-scene analysis. Its role in this work is to bridge raw LiDAR perception and higher-level trajectory optimization by supplying reliable target observations from the driving environment.

Section 2.3 Principles of Multi-Object Tracking

Multi-object tracking is an important component in autonomous driving perception, since it establishes temporal consistency for detected objects across consecutive frames. Compared with single-frame detection, tracking not only preserves object identities over time but also provides motion-related information that is essential for downstream tasks such as trajectory prediction and planning. In this work, the tracking module is built upon MCTrack [15] , which is a unified 3D multi-object tracking framework designed for autonomous driving scenarios.

MCTrack follows the tracking-by-detection paradigm. Its input consists of 3D detection results that have been converted into a unified data format, and the entire tracking pipeline is carried out in the world coordinate system. The core objective is to associate current detections with existing trajectories and maintain stable target identities over time. In this way, isolated detection results from individual frames are transformed into continuous target trajectories with interpretable motion states.

for downstream tracking and dynamic-scene analysis. Its role in this work is to bridge raw LiDAR perception and higher-level trajectory optimization by supplying reliable target observations from the driving environment.

第2.3节 多目标跟踪原理

多目标跟踪是自动驾驶感知系统中的关键组件，其核心作用在于为连续帧中检测到的物体建立时间一致性。相较于单帧检测，跟踪技术不仅能保持物体身份随时间的稳定性，还能提供运动相关数据——这些信息对轨迹预测、路径规划等下游任务至关重要。本研究基于MCTrack[15]构建了跟踪模块，该框架是专为自动驾驶场景设计的统一三维多目标跟踪系统。

MCTrack采用基于检测的跟踪范式。其输入数据为经过统一格式转换的三维检测结果，整个跟踪流程均在世界坐标系中完成。核心目标在于将实时检测结果与现有轨迹进行关联，并确保目标身份随时间保持稳定。通过这种方式，单帧检测结果被转化为具有可解释运动状态的连续目标轨迹。

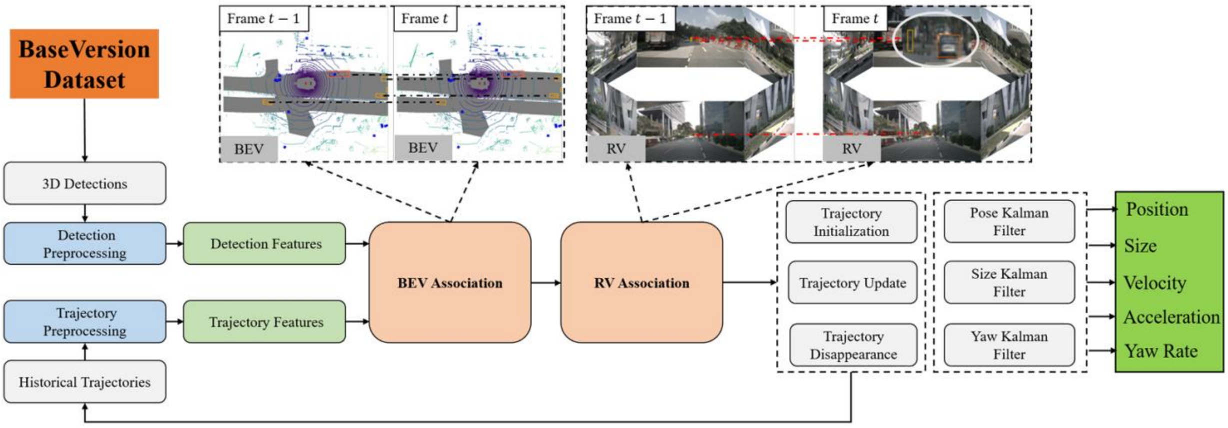

Figure 2-4 Overview of 3D MOT framework MCTrack

A key feature of MCTrack is its hierarchical association strategy. According to the framework presented in Figure 2-4, the first-stage matching is performed in the bird's-eye-view plane, where trajectory boxes and detection boxes are associated in the world coordinate system. After this primary matching step, unmatched trajectories and detections are further projected onto the image plane for secondary matching. This two-stage design improves association robustness in autonomous driving scenes, especially when targets exhibit complex motion or partial ambiguity in a single representation space.

After data association, the tracking states are updated together with the motion model. MCTrack adopts Kalman-filter-based state estimation to maintain and update the motion states of targets across frames. Through prediction and update, the tracker can preserve trajectory continuity even when detections vary over time. In addition to target identity maintenance, the framework also outputs motion-related information such as position, velocity, and acceleration, which is emphasized in the original paper as being important

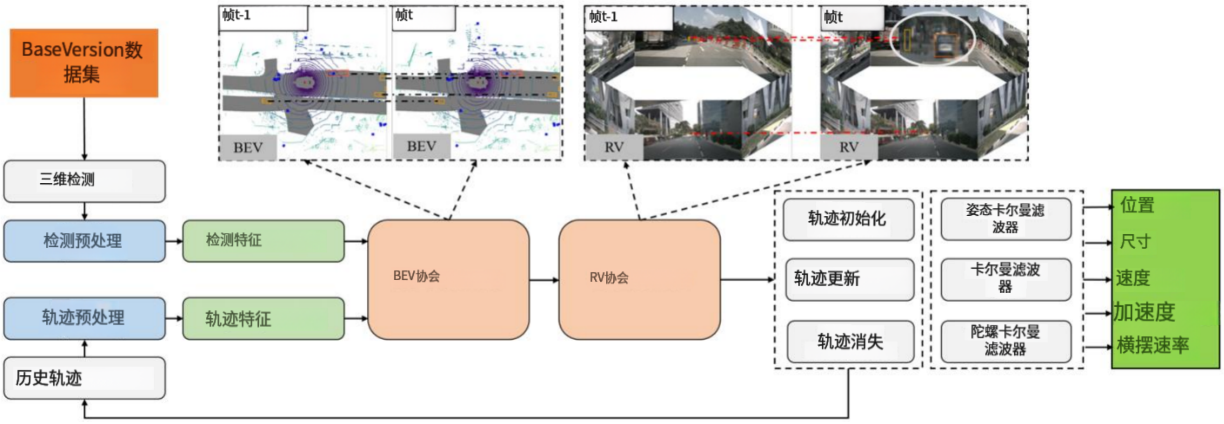

图2-4 三维运动追踪框架MCTrack概述

MCTrack的核心优势在于其分层关联策略。如图2-4所示框架结构，第一阶段匹配在鸟瞰视角平面进行，将轨迹框与检测框在世界坐标系中建立对应关系。完成初步匹配后，未匹配的轨迹和检测数据会被进一步投射到图像平面进行二次匹配。这种两阶段设计显著提升了自动驾驶场景中的关联鲁棒性，尤其在目标物体呈现复杂运动轨迹或单个表征空间存在部分模糊性时效果更为突出。

完成数据关联后，跟踪状态将与运动模型同步更新。MCTrack采用基于卡尔曼滤波的状态估计技术，实现目标运动状态在多帧间的持续维护与动态更新。通过预测与实时校正机制，即使检测结果随时间变化，跟踪器仍能保持轨迹连续性。该框架不仅具备目标身份识别能力，还能输出位置、速度、加速度等运动相关参数——这些关键信息在原始论文中被特别强调为系统的核心价值。

for downstream autonomous driving tasks.

For the present work, multi-object tracking plays a bridging role between 3D object detection and trajectory optimization. Detection provides object observations for each frame, while tracking further organizes them into temporally consistent target-level information. As a result, the system can obtain more stable dynamic-scene cues than those provided by single-frame detections alone. These tracked objects then serve as an important basis for subsequent processing in dynamic-scene trajectory optimization.

用于下游自动驾驶任务。

在本研究中，多目标跟踪技术在三维物体检测与轨迹优化之间发挥着桥梁作用。检测过程为每帧图像提供物体观测数据，而跟踪技术则将这些数据整合为时间一致性更强的目标级信息。因此，该系统能够获取比单帧检测更稳定的动态场景线索。这些被跟踪的物体随后成为动态场景轨迹优化后续处理的重要基础。

Chapter 3 System Framework and Data Flow Design

Section 3.1 Overall System Architecture

The proposed system is built as a dynamic-scene trajectory optimization framework based on FAST-LIO2. Its objective is to improve ego trajectory estimation in autonomous driving scenarios by integrating 3D object detection, multi-object tracking, front-end weight optimization, and back-end trajectory refinement into a unified pipeline. Instead of treating perception and localization as independent processes, the framework establishes explicit interactions among these modules so that dynamic-scene information can be incorporated into trajectory estimation more effectively.

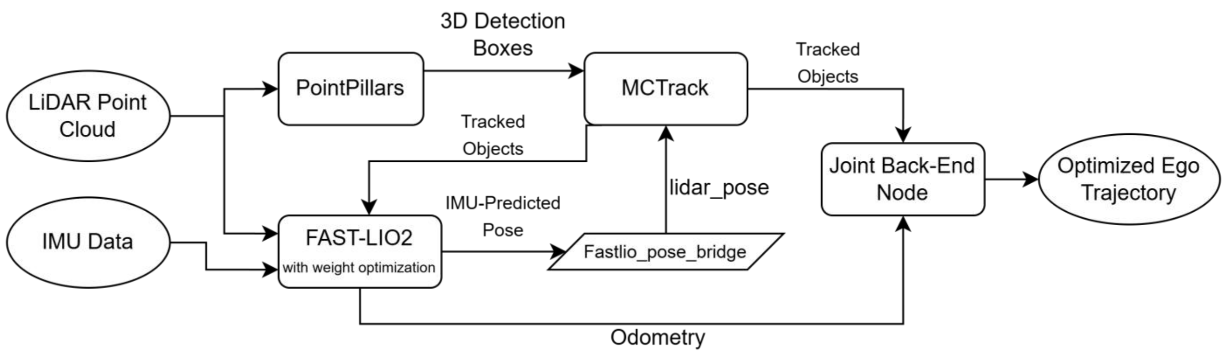

Figure 3-1 Overall architecture of the proposed dynamic-scene trajectory optimization framework

As illustrated in Figure 3-1, the system takes LiDAR point clouds and IMU data as the main inputs. The LiDAR point cloud is sent to the PointPillars module for 3D object detection, while the LiDAR and IMU measurements are

第三章 系统框架与数据流设计

第3.1节 系统整体架构

该系统基于FAST-LIO2框架构建为动态场景轨迹优化系统，其核心目标是通过将三维物体检测、多目标跟踪、前端权重优化及后端轨迹优化等模块整合到统一流程中，提升自动驾驶场景下的自我轨迹估计精度。不同于传统将感知与定位视为独立流程的做法，该框架明确建立了各模块间的交互机制，从而能够更高效地将动态场景信息融入轨迹估算过程。

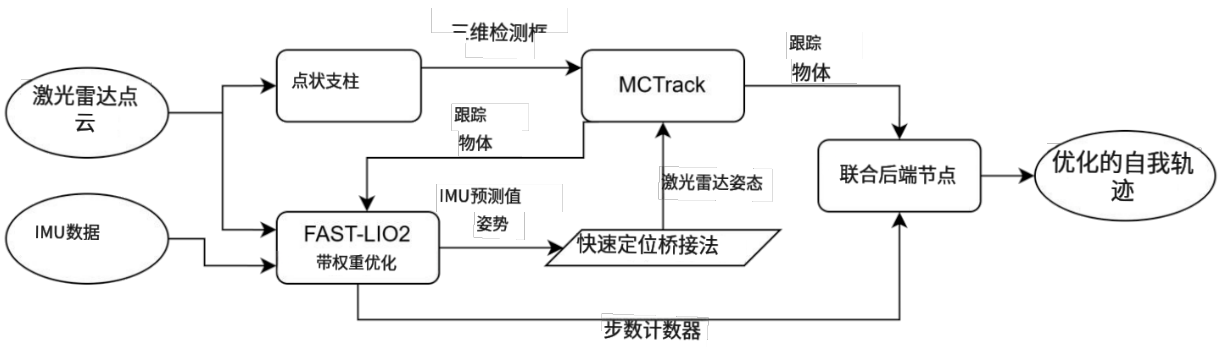

图3-1 所提动态场景轨迹优化的整体架构
构架

如图3-1所示，该系统以激光雷达点云和惯性测量单元（IMU）数据为主要输入。激光雷达点云被送入PointPillars模块进行三维目标检测，而激光雷达与IMU的测量数据则

jointly processed by FAST-LIO2 for LiDAR-inertial odometry. In this way, the framework simultaneously obtains object-level observations of the environment and ego-motion estimation of the platform from the same driving sequence.

The detection results generated by PointPillars are then forwarded to MCTrack, which performs multi-object tracking to maintain the temporal consistency of surrounding vehicles across consecutive frames. At the same time, FAST-LIO2 publishes an IMU-predicted ego pose, which is converted by the fastlio_pose_bridge module and provided to MCTrack as the pose reference for transforming detections into the global frame. Based on the combination of detection boxes and ego-pose information, MCTrack produces tracked object states that can be further used in the subsequent optimization process.

A key feature of the proposed architecture is the closed-loop interaction between MCTrack and FAST-LIO2. The tracked objects produced by MCTrack are fed back to FAST-LIO2 and used for dynamic-point weight optimization in the current odometry process. Since tracking results are generated with processing delay, the framework uses the most recently available tracked objects rather than enforcing strict same-frame feedback. This design allows dynamic-scene information to be incorporated into FAST-LIO2 in a practical and temporally tolerant manner.

In addition to the front-end feedback, the framework also includes a joint back-end node for ego trajectory refinement. This node takes the odometry

由FAST-LIO2联合处理用于激光雷达惯性里程计。通过这种方式，该框架可从同一驾驶序列中同时获取环境的物体级观测数据及平台的自我运动估计。

PointPillars生成的检测结果随后被传输至MCTrack系统，该系统通过多目标跟踪技术确保连续帧中周边车辆的时间一致性。与此同时，FAST-LIO2系统会输出IMU预测的本体位姿数据，经fastlio_pose_bridge模块转换后作为姿态参考提供给MCTrack，用于将检测结果转换至全局坐标系。基于检测框与本体位姿信息的综合分析，MCTrack生成的跟踪目标状态数据可进一步应用于后续优化流程。

该架构的核心特征在于MCTrack与FAST-LIO2之间的闭环交互机制。MCTrack生成的跟踪目标数据会反馈至FAST-LIO2系统，用于当前里程计过程中的动态点权重优化。由于跟踪结果存在处理延迟，该框架采用最新可用的跟踪目标数据，而非强制要求严格帧间同步反馈。这种设计使得动态场景信息能够以实用且时间容忍的方式被整合到FAST-LIO2系统中。

除前端反馈机制外，该框架还包含用于自我轨迹优化的联合后端节点。该节点采用里程计数据进行处理。

output of FAST-LIO2 together with tracked object observations as input, and then performs back-end processing to generate an optimized ego trajectory. As a result, the whole system contains two trajectory-oriented stages: front-end weight optimization for reducing the influence of dynamic objects during odometry estimation, and back-end refinement for further improving trajectory consistency.

Overall, the proposed architecture forms a complete pipeline that connects perception, tracking, odometry, feedback, and trajectory optimization within one system. This structure provides the foundation for the methods described in the following sections.

Section 3.2 Bidirectional Fusion Between MCTrack and FAST-LIO2

A central characteristic of the proposed framework is the bidirectional fusion between MCTrack and FAST-LIO2. Instead of operating as two isolated modules, tracking and odometry are connected through a closed-loop interaction, so that ego-motion information can support target tracking, while tracked object information can in turn support dynamic-scene trajectory optimization.

The first direction of this interaction is from FAST-LIO2 to MCTrack. During odometry estimation, FAST-LIO2 publishes an IMU-predicted ego pose before the LiDAR update step. This pose is further converted by the

系统以FAST-LIO2的输出结果及跟踪目标观测数据作为输入，通过后端处理生成优化的自我轨迹。整个系统包含两个轨迹导向阶段：前端权重优化阶段用于在里程计估算过程中降低动态目标的影响，后端精炼阶段则进一步提升轨迹一致性。

总体而言，所提出的架构形成了一套完整的流程体系，将感知、跟踪、里程计、反馈及轨迹优化等功能整合于单一系统中。该结构为后续章节所述方法提供了理论基础。

第3.2节 MCTrack与FAST-LIO2之间的双向融合

该框架的核心特征在于MCTrack与FAST-LIO2之间的双向融合机制。追踪与里程计测量并非作为独立模块运行，而是通过闭环交互机制实现联动：自我运动信息可辅助目标追踪，而被追踪物体信息则能反过来支持动态场景轨迹优化。

该相互作用的第一个作用方向是从FAST-LIO2到MCTrack。在里程计估算过程中，FAST-LIO2会在激光雷达更新步骤之前发布IMU预测的本体姿态。该姿态随后由系统进行进一步转换。

fastlio_pose_bridge module and provided to MCTrack as the ego-pose reference. In the tracking pipeline, this reference is used to transform LiDAR detections into the global frame, so that detections from consecutive frames can be represented in a consistent coordinate system. In this way, MCTrack is supported not only by frame-wise detection outputs, but also by ego-motion information from the odometry side.

The second direction of the interaction is from MCTrack back to FAST-LIO2. After data association and state update, MCTrack outputs tracked object states with temporal continuity, including object position, size, orientation, and motion-related information. These tracked objects are fed back to FAST-LIO2 and used for dynamic-region-aware processing during odometry estimation. As a result, the localization module is no longer based solely on raw geometric observations, but can also make use of structured target-level information from the tracking side.

In practice, the tracking results are produced with processing delay and therefore are not strictly synchronized with the LiDAR frame currently being processed by FAST-LIO2. For this reason, the framework uses the most recently available tracked object results rather than enforcing same-frame feedback. This design makes the closed-loop interaction feasible under practical experimental conditions while maintaining a stable data flow in the system.

fastlio_pose_bridge模块将生成的参考坐标提供给MCTrack作为本体姿态基准。在跟踪处理流程中，该基准坐标用于将激光雷达检测数据转换至全局坐标系，从而确保连续帧检测结果能在统一坐标系中呈现。通过这种机制，MCTrack不仅能接收逐帧检测输出数据，还可整合来自里程计系统的本体运动信息。

交互过程的第二个方向是从MCTrack回传至FAST-LIO2。经过数据关联与状态更新后，MCTrack会输出具有时间连续性的被跟踪物体状态信息，包括物体位置、尺寸、朝向及运动相关参数。这些被跟踪物体数据会被反馈至FAST-LIO2系统，并用于里程计估算过程中的动态区域感知处理。由此，定位模块不再仅依赖原始几何观测数据，还能有效利用来自跟踪端的结构化目标级信息。

实际应用中，跟踪结果会因处理延迟产生时延，因此与FAST-LIO2当前处理的激光雷达帧无法完全同步。为此，该框架采用最新可用的被跟踪目标结果，而非强制执行同帧反馈机制。这种设计既能在实际实验条件下实现闭环交互，又能确保系统数据流的稳定性。

Through this bidirectional fusion, FAST-LIO2 and MCTrack complement each other in the dynamic-scene pipeline. The odometry module provides ego-pose reference for tracking, while the tracking module provides temporally consistent dynamic-object information for odometry-related processing. This interaction forms the key connection between perception and localization in the proposed framework and provides the basis for the front-end weight optimization introduced in the following chapter.

Section 3.3 Joint Back-End Node for Ego Trajectory Refinement

In addition to the front-end feedback mechanism, the proposed framework further introduces a joint back-end node for ego trajectory refinement. This module is designed to improve trajectory consistency after front-end odometry estimation, especially in dynamic driving scenes where local disturbances may still remain even after dynamic-region-aware processing. Instead of replacing the original FAST-LIO2 odometry, the back-end node takes it as the primary motion estimate and performs an additional refinement process based on both odometry and tracked object information.

The input to the joint back-end node consists of two parts. The first part is the odometry output from FAST-LIO2, which provides the initial ego trajectory and inter-frame motion information. The second part is the tracked object

通过这种双向融合机制，FAST-LIO2与MCTrack在动态场景处理流程中实现了协同互补。里程计模块为跟踪任务提供本体姿态参考，而跟踪模块则为里程计相关处理提供时间一致性动态物体信息。这种交互机制构成了所提框架中感知与定位之间的关键纽带，并为后续章节将介绍的前端权重优化奠定了理论基础。

第3.3节 自主轨迹优化的联合后端节点

除前端反馈机制外，本框架还引入了用于自我轨迹优化的联合后端节点。该模块旨在提升前端里程计估算后的轨迹一致性，尤其适用于动态驾驶场景——即便经过动态区域感知处理，局部干扰仍可能残留。后端节点并未直接替换原始FAST-LIO2里程计，而是将其作为主要运动估计源，同时结合里程计数据与被追踪物体信息进行二次优化处理。

联合后端节点的输入包含两部分：第一部分为来自FAST-LIO2的里程计输出，其提供初始自我轨迹及帧间运动信息；第二部分为被跟踪目标。

information produced by MCTrack, which contains temporally continuous observations of surrounding dynamic targets. By combining these two sources, the back-end node is able to incorporate both ego-motion estimation and target-level scene information into a unified refinement process.

The core role of this node is to organize the available observations into a trajectory-oriented optimization pipeline. On the one hand, the odometry results from FAST-LIO2 provide the basic motion continuity of the ego vehicle. On the other hand, the tracked objects provide additional scene constraints from the dynamic environment. Through this design, the back-end node extends the trajectory estimation process beyond pure front-end odometry and allows the system to further improve ego-motion consistency under dynamic conditions.

The output of the joint back-end node is an optimized ego trajectory for subsequent trajectory evaluation. In this way, the proposed framework contains two complementary trajectory-oriented stages: front-end dynamic-aware processing inside FAST-LIO2, and back-end ego trajectory refinement based on odometry and tracked object observations. This layered design improves the overall robustness of the system in dynamic scenes.

Section 3.4 Semi-Synchronous Offline Frame Scheduling Mechanism

To reduce frame dropping and temporal mismatch caused by the relatively long processing time of object detection and tracking, the proposed system

MCTrack生成的信息包含对周围动态目标的连续时间观测数据。通过整合这两种数据源，后端节点能够将自我运动估计与目标级场景信息纳入统一的优化流程中。

该节点的核心功能是将现有观测数据整合为基于轨迹优化的处理流程。一方面，FAST-LIO2提供的里程计数据为自车运动提供了基础连续性保障；另一方面，被追踪目标则从动态环境中获取额外的场景约束条件。通过这种设计架构，后端节点将轨迹估计过程从单纯的前端里程计数据扩展到更全面的分析维度，使系统能够在动态环境中持续提升自车运动的一致性表现。

联合后端节点的输出是经过优化的自我轨迹，用于后续轨迹评估。该框架包含两个互补的轨迹导向阶段：FAST-LIO2内部的前端动态感知处理，以及基于里程计和被跟踪物体观测数据的后端自我轨迹精炼。这种分层设计显著提升了系统在动态场景中的整体鲁棒性。

第3.4节 半同步离线帧调度机制

为减少由目标检测与跟踪较长处理时间导致的帧丢失及时间失配问题，本系统提出

adopts a semi-synchronous offline frame scheduling mechanism during experiments. Instead of replaying all sensor data continuously at a fixed speed, the system controls the release of LiDAR frames in a more coordinated way, so that the perception branch can provide more complete results for each frame.

The basic idea is straightforward. After one LiDAR frame is released, the system does not immediately send out the next LiDAR frame. Instead, it first waits for the corresponding detection and tracking results of the current frame. In this way, the tracking branch is given enough time to process the current input, which helps reduce the situation where later modules receive missing or severely delayed object information.

However, this waiting process is not completely strict. The system mainly waits for the slower perception branch, especially detection and tracking, because these modules require much more processing time than odometry. By contrast, FAST-LIO2 and the joint back-end node are not forced to stop in the same frame-by-frame manner. Therefore, the scheduling mechanism is only semi-synchronous: it coordinates the release of LiDAR frames with perception results, but it does not fully block the entire pipeline.

A timeout strategy is also introduced to prevent the system from waiting indefinitely. If the required detection or tracking result for the current frame is not received within the allowed time, the system stops waiting and releases the next LiDAR frame. This design makes the pipeline more robust in practice,

实验中采用半同步离线帧调度机制。系统不再以固定速率连续回放全部传感器数据，而是通过更协调的方式控制激光雷达帧的释放，从而使感知分支能够为每帧提供更完整的数据结果。

其核心原理十分直观。当系统释放一个激光雷达（LiDAR）数据帧后，并不会立即发送下一帧数据，而是会先等待当前帧对应的检测与跟踪结果。这种机制为跟踪模块提供了充足的时间处理当前输入数据，从而有效避免后续模块接收到缺失或严重延迟的物体信息的情况。

不过，这种等待机制并非完全严格。系统主要会等待感知分支中较慢的环节——尤其是检测与跟踪模块，因为这些模块所需的处理时间远超里程计。相比之下，FAST-LIO2系统及其联合后端节点无需像传统方法那样逐帧强制停止。因此，调度机制仅采用半同步模式：它会协调激光雷达帧与感知结果的同步释放，但不会完全阻塞整个数据处理流程。

系统还引入了超时策略以避免无限期等待。若当前帧所需的检测或跟踪结果未在规定时间内获取，系统将停止等待并释放下一帧激光雷达数据。该设计显著提升了实际应用中的流程稳定性。

because occasional slow inference or missing outputs will not cause the whole experiment to stall.

This mechanism is mainly intended for offline experiments, where the detection and tracking modules are the major sources of delay. By controlling the LiDAR frame release in this way, the framework achieves a better balance between temporal consistency and execution efficiency, which makes the full pipeline more stable for dynamic-scene trajectory optimization experiments.

因为偶尔出现的推理速度缓慢或输出缺失不会导致整个实验停滞。

该机制主要适用于离线实验场景，其中检测与跟踪模块是延迟的主要来源。通过这种方式控制激光雷达帧的释放，该框架在时间一致性与执行效率之间实现了更优平衡，从而使整个流程在动态场景轨迹优化实验中更具稳定性。

Chapter 4 Front-End Weight Optimization Based on Tracked Dynamic Objects

Section 4.1 Dynamic Region Identification from Tracking Results

Section 4.1.1 Tracked Object Snapshot for Current-Frame Processing

In the proposed framework, dynamic-region identification for the current LiDAR frame is performed using the most recently available tracked object results. Due to the processing delay of object detection and multi-object tracking, these results are typically not strictly synchronized with the LiDAR frame currently being processed by FAST-LIO2, and in practice they usually correspond to the previous processing cycle. Before the current frame enters front-end processing, the available tracked objects are copied to form a fixed snapshot for that frame. This design ensures that all points in the same LiDAR frame are evaluated against a consistent set of tracked object states during dynamic-region identification.

The motivation for this design comes from the asynchronous nature of the perception pipeline. Since tracking results are updated at a different time from LiDAR odometry processing, directly using continuously changing target states would reduce the consistency of front-end processing. By contrast, the snapshot strategy guarantees that one LiDAR frame corresponds to one fixed set of tracked object observations, even though these observations may originate from

第4章 基于轨迹动态对象的前端权重优化

第4.1节 从追踪结果中动态识别区域

第4.1.1节 当前帧处理用跟踪对象快照

在该框架中，当前激光雷达帧的动态区域识别过程采用最新追踪目标数据进行。由于目标检测与多目标跟踪存在处理延迟，这些数据通常无法与FAST-LIO2当前处理的激光雷达帧严格同步，实际应用中往往对应前一个处理周期的数据。在当前帧进入前端处理前，系统会将可用追踪目标数据进行复制，形成该帧的固定快照。这种设计确保了在动态区域识别过程中，同一激光雷达帧内所有数据点均基于统一的追踪目标状态集合进行评估。

该设计方案的灵感源自感知流程的异步特性。由于跟踪结果的更新时间与激光雷达里程计处理存在时序差异，若直接采用持续变化的目标状态将导致前端处理一致性降低。相比之下，快照策略能确保每个激光雷达帧对应一组固定跟踪目标观测数据，即使这些观测数据可能源自不同时间点。

the immediately preceding cycle.

Each tracked object in the snapshot contains the basic geometric and temporal information required for subsequent dynamic-region identification, including the object center, box dimensions, orientation, and timestamp. These tracked objects provide structured target-level information for the current frame and serve as the basis for determining whether a LiDAR point belongs to a dynamic region.

Section 4.1.2 Expanded Dynamic Bounding Box Representation

After the tracked object snapshot is obtained for the current LiDAR frame, each tracked object is represented as an oriented three-dimensional bounding box for dynamic-region identification. A tracked object j is characterized by its box center

$$\mathbf{c}_j = [c_x^{(j)}, c_y^{(j)}, c_z^{(j)}]^{\mathrm{T}} \tag{4-1}$$

its box dimensions

$$\mathbf{d}_j = [d_x^{(j)}, d_y^{(j)}, d_z^{(j)}]^{\mathrm{T}} \tag{4-2}$$

and its yaw angle θ_j, which describes the object orientation in the horizontal plane. Based on these quantities, the tracked object can be modeled as an oriented 3D box in the global frame.

In the proposed framework, the original tracked bounding box is not used directly for dynamic-region identification. Instead, a fixed spatial margin is

紧接前一个周期

快照中的每个被追踪对象均包含后续动态区域识别所需的基础几何与时间信息，包括对象中心坐标、框尺寸、朝向及时间戳。这些被追踪对象为当前帧提供了结构化的目标级信息，并作为判定激光雷达点是否属于动态区域的基础依据。

第4.1.2节 扩展动态边界框表示法

在获取当前LiDAR帧的被跟踪对象快照后，每个被跟踪对象均以定向三维边界框形式表示，用于动态区域识别。被跟踪对象j的特征由其边界框中心点决定。

$$\mathbf{c}_j = [c_x^{(j)}, c_y^{(j)}, c_z^{(j)}]^{\mathrm{T}} \qquad (4 - 1)$$

其包装盒尺寸

$$\mathbf{d}_j = [d_x^{(j)}, d_y^{(j)}, d_z^{(j)}]^{\mathrm{T}} \qquad (4-2)$$

以及其偏航角θ_j，该角度用于描述物体在水平面内的朝向。基于这些参数，被追踪物体可在全局坐标系中建模为定向三维盒子。

在所提出的框架中，原始跟踪边界框并非直接用于动态区域识别。相反，采用固定的空间裕度

added to the box dimensions to obtain an expanded dynamic bounding box. This design is introduced to improve robustness under practical conditions, since the tracked objects used for the current LiDAR frame usually come from the most recently available tracking results rather than strictly synchronized same-frame outputs. As a result, small temporal delay, localization fluctuation, or target motion between two consecutive processing cycles may lead to spatial mismatch between the tracked box and the actual point distribution in the current frame.

To tolerate such mismatch, the half size of the expanded bounding box is defined as

$$\mathbf{h}_j = \begin{bmatrix} \frac{d_x^{(j)}}{2} + \Delta_{xy} \\ \frac{d_y^{(j)}}{2} + \Delta_{xy} \\ \frac{d_z^{(j)}}{2} + \Delta_z \end{bmatrix} \qquad (4-3)$$

Where Δ_{xy} denotes the additional margin in the horizontal plane and Δ_z denotes the additional margin in the vertical direction. In the current implementation, the horizontal margin is set to 0.6 m and the vertical margin is set to 0.5 m.

With this representation, each tracked object defines an expanded oriented spatial region around the estimated target position. Compared with using the original bounding box directly, this expanded representation provides a more

通过扩展边界框尺寸来获得动态边界框。该设计旨在提升实际应用中的鲁棒性，因为当前激光雷达帧中使用的被跟踪对象通常源自最新可用的跟踪结果，而非严格同步的同帧输出数据。因此，在连续两个处理周期之间可能出现的微小时间延迟、定位波动或目标运动，都可能导致跟踪边界框与当前帧中实际点分布之间产生空间偏差。

为容忍此类不匹配，扩展边界框的半尺寸被定义为

$$\mathbf{h}_j = \begin{bmatrix} \frac{d_x^{(j)}}{2} + \Delta_{xy} \\ \frac{d_y^{(j)}}{2} + \Delta_{xy} \\ \frac{d_z^{(j)}}{2} + \Delta_z \end{bmatrix} \tag{4-3}$$

其中Δ_{xy}表示水平方向上的附加余量，Δ_z表示垂直方向上的附加余量。

在当前实现方案中，水平余量设定为0.6米，垂直余量设定为0.5米。

采用该表示方法时，每个被追踪对象均会在估计目标位置周围定义一个扩展定向的空间区域。与直接使用原始边界框相比，这种扩展表示法提供了更精确的定位效果。

robust basis for dynamic-region identification in the front-end processing of FAST-LIO2. It allows the system to maintain tolerance to moderate delay and spatial deviation while still preserving the target-level geometric structure required for point-wise region testing. Based on the expanded dynamic bounding box, the next subsection further determines whether a LiDAR point belongs to the dynamic region of a tracked object.

Section 4.1.3 Point-in-Box Test for Dynamic Region Identification

To determine whether a LiDAR point in the current frame belongs to the dynamic region of a tracked object, a point-in-box test is performed based on the expanded dynamic bounding box. Since the tracked box is oriented by its yaw angle, this test is carried out in the local coordinate system of the object rather than directly in the global frame.

Let

$$\mathbf{p}_i = [x_i, y_i, z_i]^{\mathrm{T}} \tag{4-4}$$

denote the i-th LiDAR point in the current frame, and let c_j be the center of the j-th tracked object. The first step is to compute the relative position of the point with respect to the box center:

$$\Delta x_i^{(j)} = x_i - c_x^{(j)}, \quad \Delta y_i^{(j)} = y_i - c_y^{(j)}, \quad \Delta z_i^{(j)} = z_i - c_z^{(j)} \tag{4-5}$$

Next, the point is rotated into the local coordinate system of the tracked object according to its yaw angle θ_j. The transformed horizontal coordinates

FAST-LIO2前端处理中动态区域识别的稳健基础。该机制使系统在保持对中等延迟和空间偏差的容忍度的同时，仍能保留点级区域检测所需的目标级几何结构。基于扩展的动态边界框，下一小节进一步判定激光雷达点是否属于被追踪物体的动态区域。

第4.1.3节 动态区域识别的箱内点测试

为判断当前帧中的激光雷达点是否属于被跟踪物体的动态区域，需基于扩展动态边界框执行点框测试。由于被跟踪框的方位由偏航角决定，该测试需在物体的局部坐标系中进行，而非直接在全局坐标系中实施。

让

$$\mathbf{p}_i = [x_i,\ y_i,\ z_i]^{\mathrm{T}} \tag{4-4}$$

设当前帧中第i个LiDAR点，c_j为第j个被追踪物体的中心。第一步是计算该点相对于盒子中心的相对位置：

$$\Delta x_i^{(j)} = x_i - c_x^{(j)},\quad \Delta y_i^{(j)} = y_i - c_y^{(j)},\quad \Delta z_i^{(j)} = z_i - c_z^{(j)} \tag{4-5}$$

接下来，根据跟踪对象的偏航角θ_j，将该点旋转到跟踪对象的局部坐标系中。变换后的水平坐标

are given by

$$x_{i,\mathrm{loc}}^{(j)} = \cos\theta_j \, \Delta x_i^{(j)} + \sin\theta_j \, \Delta y_i^{(j)}, \qquad (4-6\mathrm{a})$$

$$y_{i,\mathrm{loc}}^{(j)} = -\sin\theta_j \, \Delta x_i^{(j)} + \cos\theta_j \, \Delta y_i^{(j)} \qquad (4-6\mathrm{b})$$

Based on the expanded half size

$$\mathbf{h}_j = [h_x^{(j)}, h_y^{(j)}, h_z^{(j)}]^{\mathrm{T}} \qquad (4-7)$$

the point is regarded as lying inside the dynamic region of object j if the following conditions are simultaneously satisfied:

$$\left|x_{i,\mathrm{loc}}^{(j)}\right| \le h_x^{(j)}, \quad \left|y_{i,\mathrm{loc}}^{(j)}\right| \le h_y^{(j)}, \quad \left|\Delta z_i^{(j)}\right| \le h_z^{(j)} \qquad (4-8)$$

In other words, after transforming the point into the local box frame, the system checks whether the point falls within the expanded box range along all three axes. If the above conditions hold, the point is identified as belonging to a dynamic region and will be processed differently from static background points in the subsequent weighting stage.

In addition to the geometric test, a temporal validity condition is also applied to tracked objects. Since the snapshot used by the current LiDAR frame may originate from the previous processing cycle, a tracked object is considered valid only when its timestamp is sufficiently close to the current scan time. Let t_{scan} denote the timestamp of the current LiDAR frame and t_j denote the timestamp of the j-th tracked object. Then the object is used only if

$$t_{\mathrm{scan}} - t_j \le \tau \qquad (4-9)$$

Where τ is the maximum allowable delay. In the current implementation,

由以下公式给出

$$x_i^{(\mathrm{j},\ \mathrm{l})}{}_{\mathrm{oc}} = \cos\theta_j\,\Delta x_i^{(j)} + \sin\theta_j\,\Delta y_i^{(j)}, \qquad (4-6a)$$

$$y_i^{(\mathrm{j},\ \mathrm{l})}{}_{\mathrm{oc}} = -\sin\theta_j\,\Delta x_i^{(j)} + \cos\theta_j\,\Delta y_i^{(j)} \qquad (4 — 6b)$$

基于扩展半尺寸

$$\mathbf{h}_j = [h_x^{(j)}, h_y^{(j)}, h_z^{(j)}]^{\mathrm{T}} \qquad (4-7)$$

若同时满足以下条件，则认为该点位于物体j的动态区域内：

$$\left| x_{\mathrm{i，位点}}^{(j)} \right| \le h_x^{(j)}, \quad \left| y_{i,\mathrm{loc}}^{(j)} \right| \le h_y^{(j)}, \quad \left| \Delta z_i^{(j)} \right| \le h_z^{(j)} \qquad (4-8)$$

换言之，当点被转换至局部框坐标系后，系统会检测该点是否在三个坐标轴方向上均位于扩展框区域内。若满足上述条件，则判定该点属于动态区域，并在后续加权阶段采用与静态背景点不同的处理方式。

除了几何测试外，还对被跟踪对象施加了时间有效性条件。由于当前LiDAR帧使用的快照可能源自前一处理周期，只有当被跟踪对象的时间戳与当前扫描时间足够接近时，该对象才被视为有效。设t_{scan}表示当前LiDAR帧的时间戳，t_j表示第j个被跟踪对象的时间戳，则仅当满足以下条件时该对象才会被使用：

$$t_{扫描} - t_j \le \tau \qquad (4-9)$$

其中 τ 表示最大允许延迟。在当前实现方案中，

τ is set to 1.0s. This condition prevents outdated tracking results from participating in dynamic-region identification.

Through the above point-in-box test, the current LiDAR frame is divided into points associated with tracked dynamic objects and points belonging to the remaining scene. This provides the direct basis for the adaptive weight assignment strategy described in the next section.

Section 4.2 Weight Assignment Strategy for Dynamic Points

Section 4.2.1 Object-Level Weight from Tracking Attributes

Once dynamic regions have been identified from tracked objects, the framework proceeds to assign a weight to each tracked object before point-level weighting is performed. In the proposed method, the weight of a dynamic object is determined according to its motion and tracking attributes rather than being assigned as a fixed constant. This design allows the weighting process to better reflect the reliability and dynamic intensity of different targets.

For the j-th tracked object, its motion magnitude is first represented by the speed norm and the acceleration norm:

$$v_j = \|\mathbf{v}_j\|_2, \quad a_j = \|\mathbf{a}_j\|_2 \qquad (4-10)$$

Where $\mathbf{v}_j$ and $\mathbf{a}_j$ denote the velocity vector and acceleration vector of the tracked object, respectively.

To make different attributes comparable in the weighting process, the speed, τ 设置为1.0秒。该条件可防止过时的跟踪结果参与动态区域识别过程。

通过上述点框测试，当前激光雷达帧被划分为与被追踪动态物体相关的点群及属于场景其余部分的点群。这为下一节所述的自适应权重分配策略提供了直接依据。

第4.2节 动态点权重分配策略

第4.2.1节 通过追踪属性获取对象层级权重

在通过跟踪目标识别出动态区域后，该框架会先为每个目标分配权重，再进行点级加权处理。本方法采用动态目标权重计算方式，其权重值并非固定常数，而是根据目标的运动特征和跟踪属性动态确定。这种设计使加权过程能更精准地反映不同目标的可靠性及动态强度特征。

对于第j个被追踪对象，其运动幅度首先通过速度范数和加速度范数进行表征：

$$v_j = \| \mathbf{v}_j \|_2,\ a_j = \| \mathbf{a}_j \|_2 \qquad (4-10)$$

其中 $\mathbf{v}_j$和$\mathbf{a}_j$分别表示被跟踪物体的速度向量和加速度向量。

为使加权过程中不同属性具有可比性，速度

acceleration, and detection score are normalized into the interval [0,1]:

$$\phi_v^{(j)} = \text{clip}\left(\frac{v_j}{v_{\text{ref}}}, 0{,}1\right), \quad \phi_a^{(j)} = \text{clip}\left(\frac{a_j}{a_{\text{ref}}}, 0{,}1\right), \quad \phi_s^{(j)} = \text{clip}(s_j, 0{,}1) \quad (4-11)$$

Where v_{ref} and a_{ref} are the reference speed and reference acceleration, respectively, and s_j is the detection score of the tracked object. The operator clip(·) limits the value to the range [0,1].

Based on these normalized terms, an object-level penalty is defined as

$$P_j = \lambda_v \phi_v^{(j)} + \lambda_a \phi_a^{(j)} + \lambda_s \phi_s^{(j)} \quad (4-12)$$

λ_v, λ_a and λ_s are the penalty coefficients for speed, acceleration, and score, respectively. In this formulation, a larger object speed or acceleration leads to a larger penalty, while a higher score indicates that the tracked target is more reliable and therefore its dynamic influence should be more strongly considered in the weighting process.

The final object-level weight is then defined as

$$w_j = \begin{cases} 1, & \ell_j < \ell_{\min} \\ \text{clip}(1 - P_j, w_{\min}, 1), & \ell_j \geq \ell_{\min} \end{cases} \quad (4-13)$$

Where ℓ_j denotes the track length of the j-th object, $\ell_{\min}$ is the minimum valid track length, and $w_{\min}$ is the minimum allowable dynamic weight. This formulation means that newly appeared or insufficiently stable tracked objects are not directly down-weighted. Only when the tracked object has accumulated enough temporal evidence does the system apply adaptive suppression according to its motion and confidence attributes.

加速度与检测评分均被标准化至区间[0,1]：

$$\phi_v^{(j)} = \operatorname{clip}\left(\frac{v_j}{v_{\mathrm{ref}}}，0.1\right)，\phi_a^{(j)} = \operatorname{clip}\left(\frac{a_j}{a_{\mathrm{ref}}}，0.1\right)，\phi_s^{(j)} = \operatorname{clip}(s_j，0.1) \quad (4-11)$$

其中v_{ref}和a_{ref}分别为参考速度和参考加速度，以及s_j是被跟踪对象的检测分数。操作符夹点（.）将该值限制在[0,1]范围内。

基于这些归一化项，定义对象级惩罚项为$P_j = \lambda_v \phi_v^{(j)} + \lambda_a \phi_a^{(j)} + \lambda_s \phi_s^{(j)}$

$$(4-12)$$

λ_v，λ_a和λ_s分别是速度、加速度和分数的惩罚系数。在此模型中，物体速度或加速度越大，惩罚值越高；而分数越高则表明被追踪目标更可靠，因此其动态影响应在加权过程中得到更充分的考量。

最终对象层级权重定义为

$$w_j = \begin{cases} 1, & \ell_j < \ell_{\min} \\ \operatorname{clip}(1 - P_j，w_{\min}，1)， & \ell_j \geq \ell_{\min} \end{cases} \quad (4-13)$$

其中ℓ_j表示第 j 个对象的轨迹长度，$\ell_{\min}$为最小有效轨迹长度，$w_{\min}$为最小允许动态权重。该公式意味着新出现或稳定性不足的被跟踪对象不会直接被降低权重。只有当被跟踪对象积累了足够的时间证据时，系统才会根据其运动和置信度属性进行自适应抑制。

After empirical tuning, the coefficients and reference parameters used in the current implementation are listed in Table 4-1.

Table 4-1 Weighting parameters used in object-level dynamic suppression

Parameter	Description	Value
v_{ref}	Reference speed	10.0
a_{ref}	Reference acceleration	4.0
λ_v	Speed penalty coefficient	0.5
λ_a	Acceleration penalty coefficient	0.3
λ_s	Score penalty coefficient	0.2
ℓ_{min}	Minimum track length	3
w_{min}	Minimum dynamic weight	0.2

Through this strategy, each tracked object is assigned a dynamic-aware weight before point-level processing. The resulting object-level weight is then propagated to the points falling inside its dynamic region, as described in the next subsection.

Section 4.2.2 Point-Level Weight Assignment

After the object-level weight w_j is determined for each tracked object, the next step is to assign a weight to each LiDAR point in the current frame. In the proposed framework, point-level weighting is performed according to the spatial relationship between the point and the expanded dynamic regions identified in Section 4.1. If a point does not belong to any dynamic region, it is

经过经验性调校后，当前实施方案中使用的系数及参考参数列于表4-1。

表4-1 对象级动态抑制中使用的加权参数

参数	描述	值
$v_{反射}$	参考速度	10.0
$a_{反射}$	参考加速度	4.0
λ_v	速度惩罚系数	0.5
λ_a	加速度惩罚系数	0.3
λ_s	评分惩罚系数	0.2
$\ell_{\min}$	最小轨道长度	3
$w_{分}$	最小动态重量	0.2

通过该策略，在点级处理前为每个被追踪对象分配动态感知权重。随后，如下一小节所述，将生成的对象级权重传播至其动态区域内所有点。

第4.2.2节 点级权重分配

在确定每个被跟踪对象的物体级权重w_j后，下一步需要为当前帧中的每个激光雷达点分配权重。在本框架中，点级权重分配依据点与第4.1节所识别的扩展动态区域之间的空间关系进行。若某点不属于任何动态区域，则

treated as a static-background point and keeps the default weight. Otherwise, its weight is inherited from the associated tracked object. When a point lies inside multiple dynamic bounding boxes, the smallest object-level weight is used.

Accordingly, the point-level weight is defined as

$$w_i = \begin{cases} 1, & p_i \notin \bigcup_j \mathcal{B}_j \\ \min_{j: p_i \in \mathcal{B}_j} w_j, & p_i \in \bigcup_j \mathcal{B}_j \end{cases} \qquad (4-14)$$

Where $\mathcal{B}_j$ denotes the expanded dynamic bounding box of the j-th tracked object.

This strategy does not directly remove dynamic points from the current LiDAR frame. Instead, dynamic points are preserved but assigned smaller weights according to the properties of the associated tracked objects. In this way, the framework reduces the influence of unstable observations while maintaining the continuity of front-end point cloud processing. These point-level weights are then incorporated into the FAST-LIO2 front end, as described in the next section.

Section 4.3 Integration of Dynamic Weighting into FAST-LIO2

The dynamic weighting strategy is integrated into the FAST-LIO2 front end rather than implemented as an external filtering module. In the proposed framework, dynamic-region identification and weight assignment are first

该点被视为静态背景点并保持默认权重值。否则，其权重将继承自关联的跟踪对象。当某点位于多个动态边界框内时，将采用最小的对象层级权重值。

因此，点水平权重被定义为

$$w_i = \begin{cases} 1, & p_i \in \bigcup_j \mathcal{B}_j \\ \underset{j:p_i \in \mathcal{B}_j}{分} w_j, & \pi \in \bigcup_j \mathcal{B}_j \end{cases} \qquad (4-14)$$

其中 B $_j$表示第 j 个被跟踪对象的扩展动态边界框。

该策略不会直接从当前激光雷达坐标系中剔除动态点，而是保留动态点并根据关联跟踪对象的特性为其分配较小权重。通过这种方式，框架在保持前端点云处理连续性的同时，有效降低了不稳定观测值的影响。随后如下一节所述，这些点级权重将被整合至FAST-LIO2前端系统中。

第4.3节 动态加权技术在FAST-LIO2中的整合

动态加权策略被集成到FAST-LIO2前端系统中，而非作为外部滤波模块实现。在所提出的框架中，动态区域识别与权重分配首先

completed during point preprocessing, and the resulting point-level weights are then directly carried into the subsequent odometry estimation process. In this way, dynamic-object information is incorporated into the front-end pipeline without changing the overall structure of FAST-LIO2.

At the preprocessing stage, each LiDAR point in the current frame is evaluated using the tracked object snapshot described in Section 4.1. According to the point-level weighting rule in Section 4.2, a dynamic weight is assigned to the point and written into its intensity field. In the current implementation, the dynamic weight is combined with the original point intensity by taking their minimum value, so that the influence of points in dynamic regions can be effectively suppressed before front-end optimization starts. Therefore, when the current frame enters the FAST-LIO2 front end, the point cloud is no longer treated as an unweighted raw observation set, but as a dynamically weighted point set whose effective weights are stored in the point attributes.

During odometry estimation, these point-level weights are further integrated into the measurement model of FAST-LIO2. Let I_i denote the weight stored in the intensity field of the i-th point during preprocessing. The effective front-end weight is written as

$$w_i = \max(w_{\min}, \min(1, I_i)) \qquad (4-15)$$

Where $w_{\min}$ is the minimum allowable dynamic weight. Based on this weight, the measurement Jacobian and residual of the corresponding point are

该过程在点数据预处理阶段完成，生成的点级权重随后直接应用于后续里程计估算流程。通过这种方式，动态目标信息得以整合至前端处理流程中，且无需改变FAST-LIO2的整体架构。

在预处理阶段，系统会根据第4.1节所述的跟踪目标快照对当前帧中的每个激光雷达点进行评估。依据第4.2节定义的点级加权规则，系统会为每个点分配动态权重并写入其强度场。当前实现方案中，动态权重会与原始点强度值取最小值进行组合处理，从而在前端优化开始前有效抑制动态区域中点数据的影响。因此当当前帧进入FAST-LIO2前端处理时，点云数据不再被视为未经加权的原始观测集，而是作为动态加权点集处理——其有效权重已存储在点属性中。

在里程计估算过程中，这些点级权重会被进一步整合到FAST-LIO2的测量模型中。设I_i表示预处理阶段存储在第i个点强度场中的权重值，其有效前端权重可表示为

$$w_i = \max(w_{\min}，\min(1，I_i)) \qquad (4-15)$$

其中$w_{\min}$表示允许的最小动态重量。基于该重量，测量点的雅可比和残差

scaled as

$$\mathbf{H}_i^{'} = w_i \mathbf{H}_i \qquad (4-16)$$

$$r_i^{'} = w_i r_i \qquad (4-17)$$

$\mathbf{H}_i$ and r_i denote the original measurement Jacobian and residual of the i-th point, and $\mathbf{H}_i^{'}$ and $r_i^{'}$ denote their weighted forms. As a result, points associated with dynamic objects contribute less to the state update than static-background points. This allows the front end to suppress unstable observations while preserving the continuity of the original optimization process.

An important characteristic of this design is that dynamic points are not directly removed from the scan. Instead, they remain in the point cloud but with reduced influence in registration. Compared with hard removal, this strategy is more suitable for the current FAST-LIO2 framework because it preserves the basic geometric structure of the scan while still reducing the negative impact of moving objects. In this sense, the proposed method introduces dynamic awareness into the front-end estimation process in a soft and continuous manner.

In addition to odometry estimation, the weighting strategy also influences map quality. Points that have been heavily down-weighted are further suppressed during map saving, which reduces the possibility that strong dynamic interference is written into the final map. This helps alleviate dynamic

按比例缩放

$$\mathbf{H}_i^{'}=w_i\mathbf{H}_i \tag{4-16}$$

$$r_{我}^{'}=w_i r_i \tag{4-17}$$

$\mathbf{H}_i$和r_i分别表示原始测量值雅可比和残差

第i个点，$\mathbf{H}_i^{'}$和$r_i^{'}$分别表示它们的加权形式。因此，这些点

与动态对象相关的数据对状态更新的贡献度低于静态背景点。这使得前

端能够抑制不稳定观测值，同时保持原始优化过程的连续性。

该设计方案的核心特征在于动态点不会被直接从扫描数据中剔除。这些点仍保留在点云中，但在配准过程中影响力会降低。相较于硬性剔除方法，这种策略更符合当前FAST-LIO2框架的需求——既能保留扫描数据的基本几何结构，又能有效减少移动物体带来的负面影响。从这个角度来看，所提出的方法以软性且连续的方式将动态感知机制融入前端估计流程。

除里程计估算外，加权策略还会对地图质量产生影响。在地图保存过程中，经过大幅降权处理的点位会进一步被抑制，从而降低强动态干扰被写入最终地图的可能性。这有助于缓解动态干扰。

ghosting and makes the resulting map cleaner in dynamic scenes. Therefore, the proposed front-end weighting strategy improves not only the robustness of odometry estimation, but also the quality of map construction under dynamic conditions.

该方法能有效消除虚化现象，并使动态场景下的地图生成结果更加清晰。因此，所提出的前端加权策略不仅提升了里程计估计的鲁棒性，还显著改善了动态条件下地图构建的质量。

Chapter 5 Back-End Joint Optimization for Ego Trajectory Refinement

Section 5.1 Odometry Consistency Constraint

The back-end optimization is performed in a sliding-window manner, where the optimization target is the ego trajectory within the current window. Let the ego state at time step k be defined as

$$\mathbf{x}_k = \begin{bmatrix} \mathbf{t}_k \\ \psi_k \end{bmatrix}, \quad \mathbf{t}_k = [x_k, y_k, z_k]^{\mathrm{T}} \qquad (5-1)$$

Where $\mathbf{t}_k$ denotes the translation of the ego vehicle and ψ_k denotes its yaw angle. Accordingly, the optimization variable in the window is written as

$$\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_N\} \qquad (5-2)$$

The first type of constraint in the back-end optimization is the odometry consistency constraint. Its purpose is to preserve the motion continuity provided by FAST-LIO2 and to prevent the refined trajectory from deviating excessively from the original front-end estimation. Since the front-end odometry already provides a reliable initial trajectory, the back-end optimization is not designed to replace it, but to refine it while maintaining inter-frame consistency.

For two adjacent ego states $\mathbf{x}_k$ and $\mathbf{x}_{k+1}$, the corresponding FAST-LIO2 output provides a relative motion measurement between the two frames. Let the measured relative translation and yaw increment be denoted by

第5章 自主轨迹优化的后端联合优化

第5.1节 距步计一致性约束

后端优化采用滑动窗口方式进行，其中优化目标为当前窗口内的自我轨迹。令时间步k处的自我状态定义为

$$\mathbf{x}_k = \begin{bmatrix} \text{托克劳群岛} \\ \psi_k \end{bmatrix}, \quad \mathbf{t}_k = [x_k, y_k, z_k]^{\mathrm{T}} \qquad (5-1)$$

其中$\mathbf{t}_k$表示自我车辆的平移量，ψ_k表示其

偏航角。因此，窗口中的优化变量表示为$\mathbf{X} = \{\mathbf{x}_1，\mathbf{x}_2，\cdots，\mathrm{x}_N\}$ $(5-2)$

后端优化中的第一类约束条件是里程计一致性约束。其核心目标在于保持FAST-LIO2提供的运动连续性，并防止优化后的轨迹与原始前端估计结果产生显著偏差。由于前端里程计已提供可靠的初始轨迹数据，后端优化并非旨在替代该过程，而是通过在保持帧间一致性的同时对初始轨迹进行精细化处理。

对于两个相邻的自我状态$\mathbf{x}_k$和$\mathbf{x}_{k+1}$，对应的FAST-LIO2输出提供了两帧之间的相对运动测量值。设测得的相对平移量和偏航增量分别表示为

$$\hat{\mathbf{u}}_k = \begin{bmatrix} \hat{\mathbf{d}}_k \\ \widehat{\Delta\psi}_k \end{bmatrix}, \quad \hat{\mathbf{d}}_k = \begin{bmatrix} \hat{d}_{x,k} \\ \hat{d}_{y,k} \\ \hat{d}_{z,k} \end{bmatrix} \tag{5-3}$$

Where $\hat{\mathbf{d}}_k$ is the relative translation from frame k to frame k+1, and $\widehat{\Delta\psi}_k$ is the corresponding yaw increment.

Based on the optimized states, the estimated relative motion between two adjacent frames can be written as

$$\mathbf{u}_k(\mathbf{X}) = \begin{bmatrix} \mathbf{t}_{k+1} - \mathbf{t}_k \\ \psi_{k+1} - \psi_k \end{bmatrix} \tag{5-4}$$

Accordingly, the odometry consistency residual is defined as

$$\mathbf{r}_{\text{odom}}^{(k)} = \begin{bmatrix} \mathbf{t}_{k+1} - \mathbf{t}_k - \hat{\mathbf{d}}_k \\ \psi_{k+1} - \psi_k - \widehat{\Delta\psi}_k \end{bmatrix} \tag{5-5}$$

This residual measures the deviation between the optimized inter-frame motion and the odometry measurement provided by FAST-LIO2. When the residual is small, the refined trajectory remains close to the original front-end odometry; when the residual becomes large, the optimization is penalized accordingly. Therefore, this term serves as the basic prior of the back-end formulation and maintains the smoothness and continuity of the ego trajectory within the sliding window.

For a window containing N ego states, the odometry consistency term can be accumulated over all adjacent frame pairs:

$$E_{\text{odom}}(\mathbf{X}) = \sum_{k=1}^{N-1} \left\| \mathbf{r}_{\text{odom}}^{(k)} \right\|^2 \tag{5-6}$$

$$\hat{\mathbf{u}}_k = \begin{bmatrix} \hat{\mathbf{d}}_k \\ \widehat{\Delta\psi}_k \end{bmatrix}, \quad \hat{\mathbf{d}}_k = \begin{bmatrix} \hat{d}_{x,k} \\ \hat{d}_{y,k} \\ \hat{d}_{z,k} \end{bmatrix} \tag{5-3}$$

其中$\hat{\mathbf{d}}_k$表示从帧k到帧k+1的相对平移量，$\widehat{\Delta\psi}_k$为相应的偏航增量。

基于优化状态，估算两个物体之间的相对运动相邻帧可表示为

$$\mathbf{u}_k(\mathbf{X}) = \begin{bmatrix} \mathbf{t}_{k+1} - \mathbf{t}_k \\ \psi_{k+1} - \psi_k \end{bmatrix} \tag{5-4}$$

因此，里程计一致性残差被定义为

$$\mathbf{r}_{\text{odom}}^{(k)} = \begin{bmatrix} \mathbf{t}_{k+1} - \mathbf{t}_k - \hat{\mathbf{d}}_k \\ \psi_{k+1} - \psi_k - \widehat{\Delta\psi}_k \end{bmatrix} \tag{5-5}$$

该残差用于衡量优化后的帧间运动与FAST-LIO2提供的里程计测量值之间的偏差。当残差较小时，优化后的轨迹仍接近原始前端里程计数据；当残差增大时，优化过程会相应受到惩罚。因此，该项作为后端公式的基本先验条件，确保了滑动窗口内本体轨迹的平滑性与连续性。

对于包含N个自我状态的窗口，里程计一致性项可对所有相邻帧对进行累积：

$$E_{\text{odom}}(\mathbf{X}) = \sum_{k=1}^{N-1} \left\| \mathbf{r}_{\text{odom}}^{(k)} \right\|^2 \tag{5-6}$$

Through this formulation, the back-end optimization preserves the fundamental motion structure estimated by FAST-LIO2 while leaving room for additional object-based observations to further refine the trajectory. The next section introduces the object observation consistency constraint, which provides complementary dynamic-scene information for ego trajectory correction.

Section 5.2 Object Observation Consistency Constraint

In addition to odometry consistency, the back-end optimization also incorporates object observation consistency as an additional source of trajectory correction. The basic idea is that tracked objects provide temporally continuous observations of the dynamic scene, and these observations can be used to constrain the ego trajectory from another perspective. If the ego poses in the sliding window are accurate, the object observations associated with these poses should remain geometrically consistent after coordinate transformation.

For each ego state $\mathbf{x}_k$, the system searches for the temporally matched tracked-object message and extracts valid tracked objects from it. Let the j-th tracked object associated with state k be denoted by its observed position and heading in the global frame:

该公式设计使得后端优化既保留了FAST-LIO2算法估算的基本运动结构，又为后续基于物体的观测数据留有空间以进一步优化轨迹。下一节将介绍物体观测一致性约束条件，该约束为自我轨迹校正提供了互补的动态场景信息。

第5.2节 对象观察一致性约束

除了里程计一致性外，后端优化还引入了物体观测一致性作为轨迹校正的补充来源。其核心原理在于：被追踪物体能提供动态场景中时间连续的观测数据，这些观测数据可从不同角度约束自我轨迹的确定。若滑动窗口中的自我位姿保持精准，那么与这些位姿相关的物体观测数据在坐标变换后仍应保持几何一致性。

对于每个自我状态$\mathbf{x}_k$，系统会搜索时间匹配的被跟踪对象信息，并从中提取有效跟踪对象。将状态k对应的第j个被跟踪对象表示为其在全局坐标系中的观测位置与航向：

$$\hat{\mathbf{o}}_{k,j} = \begin{bmatrix} \hat{\mathbf{p}}_{k,j} \\ \hat{\psi}_{k,j} \end{bmatrix}, \quad \hat{\mathbf{p}}_{k,j} = \begin{bmatrix} \hat{x}_{k,j} \\ \hat{y}_{k,j} \end{bmatrix} \tag{5-7}$$

Where $\hat{\mathbf{p}}_{k,j}$ is the tracked object position on the horizontal plane and $\hat{\psi}_{k,j}$ is the corresponding object heading.

At the same time, for each tracked object, the system also retains its relative observation with respect to the ego vehicle. Let this local observation be denoted by

$$\mathbf{z}_{k,j}^{\mathrm{loc}} = \begin{bmatrix} \mathbf{p}_{k,j}^{\mathrm{loc}} \\ \psi_{k,j}^{\mathrm{loc}} \end{bmatrix}, \quad \mathbf{p}_{k,j}^{\mathrm{loc}} = \begin{bmatrix} x_{k,j}^{\mathrm{loc}} \\ y_{k,j}^{\mathrm{loc}} \end{bmatrix} \tag{5-8}$$

$\mathbf{p}_{k,j}^{\mathrm{loc}}$ is the object position in the local frame of the ego vehicle and $\psi_{k,j}^{\mathrm{loc}}$ is the relative heading observation.

Given the ego pose $\mathbf{x}_k$, the corresponding object position predicted in the global frame can be written as

$$\mathbf{p}_{k,j}^{\mathrm{pred}} = \mathbf{R}(\psi_k)\mathbf{p}_{k,j}^{\mathrm{loc}} + \mathbf{t}_k^{xy} \tag{5-9}$$

Where

$$\mathbf{t}_k^{xy} = \begin{bmatrix} x_k \\ y_k \end{bmatrix}, \quad \mathbf{R}(\psi_k) = \begin{bmatrix} \cos\psi_k & -\sin\psi_k \\ \sin\psi_k & \cos\psi_k \end{bmatrix} \tag{5-10}$$

Similarly, the predicted global heading of the tracked object is given by

$$\psi_{k,j}^{\mathrm{pred}} = \psi_k + \psi_{k,j}^{\mathrm{loc}} \tag{5-11}$$

Based on these predicted quantities, the object observation consistency

$\hat{\mathbf{p}}_{k,j}$ $\hat{x}_{k,j}$

$$\hat{\mathbf{o}}_{k,j} = \begin{bmatrix} \\ \hat{\psi}_{k,j} \end{bmatrix}, \quad \hat{\mathbf{p}}_{k,j} = \begin{bmatrix} \\ \hat{y}_{k,j} \end{bmatrix} \qquad (5-7)$$

在哪里$\hat{\mathbf{p}}_k$、j表示水平面上被追踪物体的位置，ψ_k、j
是对应对象的标题。

与此同时，对于每个被追踪对象，系统还会保留其相对于自车的相对观测数据。该局部观测数据用以下符号表示：

通信线路 通信线路

$$\mathbf{z}^{\text{loc}}_{k,j} = \begin{bmatrix} \mathbf{p}_{k,j} \\ \psi^{\text{loc}}_{k,j} \end{bmatrix}, \quad \mathbf{p}^{\text{loc}}_{k,j} = \begin{bmatrix} x_{k,j} \\ y^{\text{loc}}_{k,j} \end{bmatrix} \qquad (5-8)$$

plok中，cj表示自我车辆局部坐标系中的物体位置，ψ lok、cj表示相对航向观测值。

给定自我姿态$\mathbf{x}_k$，全局坐标系中预测的对应物体位置可表示为

ppkr，j ed=$\mathbf{R}(\psi_k)$plok，cj +txk ^{y} $(5-9)$

在哪里

$$\mathbf{t}_k^{xy} = \begin{bmatrix} x_k \\ \text{YK} \end{bmatrix}, \quad (\psi_k) = \quad \begin{bmatrix} \cos\psi_k \ \mathbf{R} & -\sin\psi_k \\ \sin\psi_k & \cos\psi_k \end{bmatrix} \qquad (5-10)$$

同样地，被跟踪对象的预测全局航向由公式 ψ pkr，j ed=ψ_k+ ψ lok，cj给出。

$(5-11)$

基于这些预测量，物体观测一致性

residual is defined as

$$\mathbf{r}_{\text{obj}}^{(k,j)} = \begin{bmatrix} \mathbf{p}_{k,j}^{\text{pred}} - \hat{\mathbf{p}}_{k,j} \\ \text{wrap}\left(\psi_{k,j}^{\text{pred}} - \hat{\psi}_{k,j}\right) \end{bmatrix} \tag{5-12}$$

Where wrap(·) denotes angle normalization to a principal interval such as $(-\pi, \pi]$. The first two components of the residual measure the position inconsistency of the tracked object in the global frame, while the last component measures the heading inconsistency.

The object observation term over the entire sliding window can then be accumulated as

$$E_{\text{obj}}(\mathbf{X}) = \sum_{k} \sum_{j \in \mathcal{O}_k} \left\| \mathbf{r}_{\text{obj}}^{(k,j)} \right\|^2 \tag{5-13}$$

$\mathcal{O}_k$ denotes the set of valid tracked objects associated with state k.

Compared with the odometry consistency term, this object observation consistency term provides complementary information from the dynamic scene. It does not replace the front-end trajectory prior, but introduces additional geometric constraints through tracked-object observations. In this way, the back-end optimization can further correct ego trajectory deviation when odometry alone is insufficient to fully suppress the influence of dynamic interference.

残差定义为

$$\mathbf{r}_{\text{物体}}^{(k,j)} = \begin{bmatrix} \mathbf{p}_{k,j}^{\text{pred}} - \widehat{\mathbf{p}}_{k,j} \\ \text{包}\left(\psi_{k,j}^{\text{pred}} - \widehat{\psi}_{k,j}\right) \end{bmatrix} \qquad (5-12)$$

其中wrap（.）表示角度归一化至主区间（如(— π，π]。残差的前两个分量用于衡量被跟踪对象在全局坐标系中的位置不一致性，最后一个分量则用于衡量航向不一致性。

随后可将整个滑动窗口内的目标观测值项进行累积

$$E_{\text{obj}}(\mathbf{X}) = \sum_{k} \sum_{j \in \mathcal{O}_k} \left\| \mathbf{r}_{\text{obj}}^{(k,j)} \right\|^2 \qquad (5-13)$$

$\mathcal{O}_k$表示与状态 k 相关联的有效跟踪对象集合。

与里程计一致性项相比，该目标观测一致性项能从动态场景中提供补充信息。它并未替代前端轨迹先验，而是通过被追踪目标的观测数据引入额外几何约束。当仅依靠里程计数据不足以完全抑制动态干扰影响时，后端优化过程可进一步校正自我轨迹偏差。

Section 5.3 Joint Objective Function and Optimization Parameters

Based on the odometry consistency term and the object observation consistency term defined above, the back-end refinement is formulated as a joint optimization problem over the ego trajectory in the sliding window. The objective is to preserve the continuity of the original FAST-LIO2 odometry while introducing additional corrections from tracked object observations. In this way, the refined trajectory is constrained not only by inter-frame ego motion but also by the geometric consistency of dynamic-scene observations.

In the current implementation, the odometry term serves as the basic prior of the optimization, while the object observation term is additionally controlled by a global weighting factor. Accordingly, the overall objective can be written as

$$E(\mathbf{X}) = E_{\mathrm{odom}}(\mathbf{X}) + \lambda_{\mathrm{obj}} E_{\mathrm{obj}}(\mathbf{X}) \qquad (5-14)$$

Where λ_{obj} controls the influence of the object observation term relative to the odometry term.

In the practical implementation, each object residual is further weighted according to the reliability of the corresponding tracked object. Let $s_{k,j}$ be the score of object j at state k, and let $\ell_{k,j}$ be its track length. These two quantities are normalized as

第5.3节 共同目标函数与优化参数

基于前文定义的里程计一致性项与目标观测一致性项，后端优化问题被构建为滑动窗口内自我轨迹的联合优化模型。其核心目标是在保留原始FAST-LIO2里程计连续性的同时，通过跟踪目标观测数据引入额外校正。通过这种优化方式，精化轨迹不仅受帧间自我运动约束，还需满足动态场景观测数据的几何一致性要求。

在当前实现方案中，里程计术语作为优化过程的基本先验条件，而目标观测项则额外受到全局权重因子的调控。因此，整体目标函数可表述为

$$E(\mathbf{X}) = E_{\mathrm{odom}}(\mathbf{X}) + \lambda_{\mathrm{obj}} E_{\mathrm{obj}}(\mathbf{X}) \tag{5-14}$$

其中λ_{obj}用于控制物体观测项相对于里程计项的影响。

在实际实现中，每个目标残差会根据对应被跟踪目标的可靠性进一步加权。设s_k、$_{\mathrm{j}}$为状态 k 下目标 j 的得分，令$\{_k$、$_{\mathrm{j}}$为其跟踪长度。这两个量被归一化为

$$\bar{s}_{k,j} = \mathrm{clip}(s_{k,j}, 0,1), \quad \bar{\ell}_{k,j} = \mathrm{clip}\left(\frac{\ell_{k,j} - \ell_{\min} \quad + 1}{\ell_{\mathrm{ref}}}, 0,1\right) \qquad (5-15)$$

Where $\ell_{\min}$ is the minimum valid track length and ℓ_{ref} is the normalization range for track length. In the current code, ℓ_{ref} is implemented as a fixed constant of 6.0 after threshold shifting.

The weight assigned to the corresponding object residual is then written as

$$w_{k,j}^{\mathrm{obj}} = \max\left(w_{\mathrm{obj,min}}, \sqrt{\lambda_{\mathrm{obj}}}\left(0.5\bar{s}_{k,j} + 0.5\bar{\ell}_{k,j}\right)\right) \qquad (5-16)$$

Where $w_{\mathrm{obj,min}}$ is the minimum object residual weight. This formulation means that high-confidence and long-lived tracks exert stronger influence in the object consistency term, while short or less reliable tracks contribute less to the optimization.

Accordingly, the joint optimization problem can be expressed as

$$\mathbf{X}^* = arg \min_{\mathbf{X}} \left[E_{\mathrm{odom}}(\mathbf{X}) + \sum_{k} \sum_{j \in \mathcal{O}_k} \rho\left(\left\| w_{k,j}^{\mathrm{obj}} \mathbf{r}_{\mathrm{obj}}^{(k,j)} \right\|^2\right)\right] \qquad (5-17)$$

Where $\rho(\cdot)$ is the robust loss function used in the optimization. In the current implementation, Huber loss is adopted by default.

The optimization-related parameters were tuned empirically during development, and the final settings used in the present implementation are listed in Table 5-1.

$$\bar{s}_{k,j}=\mathrm{clip}(s_{k,j},0,1),\quad \bar{\ell}_{k,j}=\mathrm{clip}\left(\frac{\ell_{k,j}-\ell_{\min}+1}{\ell_{\mathrm{ref}}},0,1\right) \quad (5-15)$$

其中$\ell_{\min}$表示最小有效轨迹长度，ℓ_{ref}表示轨迹长度的归一化范围。在当前代码中，ℓ_{ref}在阈值偏移后被实现为固定常数6.0。

分配给相应对象残差的权重随后表示为

$$w^{\mathrm{obj}}_{k,j}=\max\left(w_{\mathrm{obj},\min},\ \sqrt{\lambda_{\mathrm{obj}}}\left(0.5\bar{s}_{k,j}+0.5\bar{\ell}_{k,j}\right)\right) \qquad (5-16)$$

其中 $w_{\mathrm{obj},\min}$表示物体残余权重的最小值。该公式表明：高置信度且持续时间较长的轨迹对物体一致性项的影响更强，而短时或可靠性较低的轨迹对优化过程的贡献较小。

因此，联合优化问题可表述为

$$\mathbf{X}^{*}=arg\min_{\mathbf{X}}\left[E_{\mathrm{odom}}(\mathbf{X})+\sum_{k}\sum_{j\in\mathcal{O}_k}\rho\left(\left\| w^{\mathrm{obj}}_{k,j}\,\mathbf{r}^{(k,j)}_{\mathrm{obj}}\right\|^{2}\right)\right] \qquad (5-17)$$

其中ρ（.）表示优化过程中采用的稳健损失函数。在当前实现中，默认采用Huber损失函数。

优化相关参数在开发过程中通过经验调参确定，本实施版本采用的最终参数设置列于表5-1。

Table 5-1 Optimization-related parameters used in the joint back-end module

Parameter	Description	Value
N	Sliding-window length	8
λ_{obj}	Global weight for object observation term	0.12
$w_{\text{obj,min}}$	Minimum object residual weight	0.03
ℓ_{ref}	Track-length normalization range	6.0
Robust loss	Loss function for nonlinear optimization	Huber
f_{scale}	Robust loss scale	0.5
σ_{odom}	Fallback odometry noise$[x, y, \psi]$	[0.20, 0.20, 0.10]

This formulation combines the continuity of front-end odometry with the corrective information provided by tracked objects. As a result, the optimized trajectory remains close to the original ego-motion estimate while becoming more consistent with dynamic-scene observations in the sliding window.

Section 5.4 Valid Target Gating and Optimization Procedure

In practical dynamic scenes, tracked object observations may contain noise, unstable tracks, or temporal mismatches. Directly incorporating all tracked objects into the optimization can degrade performance and introduce additional errors. Therefore, a target gating mechanism is applied before constructing the object observation constraints, so that only reliable and temporally consistent objects are used in the back-end refinement.

The gating process first evaluates each tracked object based on its

表5-1 联合后端模块中使用的优化相关参数

参数	描述	值
N	滑动窗口长度	8
λ_{obj}	物体观测项的全局权重	0.12
$w_{\mathrm{obj,min}}$	最小物体残留重量	0.03
ℓ_{ref}	曲目长度标准化范围	6.0
强健损失	非线性优化中的损失函数	休伯
法斯卡	强健损失尺度	0.5
$\sigma_{\text{距离计}}$	回退式里程计噪声[x，y，ψ]	[0.20, 0.20, 0.10]

该算法将前端里程计的连续性与被追踪物体提供的校正信息相结合。因此，优化后的轨迹在保持接近原始自我运动估计值的同时，能更好地与滑动窗口中的动态场景观测结果保持一致性。

第5.4节 有效靶向门控与优化程序

在实际动态场景中，被追踪目标的观测数据可能存在噪声干扰、轨迹不稳定或时间失配等问题。若直接将所有被追踪目标纳入优化过程，不仅会降低系统性能，还会引入额外误差。为此，我们在构建目标观测约束条件前采用目标门控机制，确保后端优化阶段仅使用可靠且时间一致性良好的目标数据。

门控过程首先根据每个被追踪对象的特征进行评估

confidence score. Only objects whose scores exceed a predefined threshold are considered for further processing. This step removes low-confidence detections that are likely to be false positives or poorly localized.

Track stability is then assessed using the track length. Objects with very short trajectories are excluded, since they usually correspond to newly initialized or unstable tracks. By enforcing a minimum track length requirement, the system favors objects with sufficient temporal consistency.

In addition, a motion-consistency constraint is introduced to filter out abnormal targets. Objects with excessively large velocity changes are rejected. Such cases often indicate tracking failure, identity switches, or incorrect motion estimation. By removing these outliers, the remaining object observations become more reliable for trajectory refinement.

Temporal alignment between ego states and tracked objects is also considered. For each state in the sliding window, the system searches for the closest tracked-object message within a limited time interval. If no suitable match is found within the allowed time difference, the corresponding object observations are discarded. This avoids introducing spatial inconsistency caused by time misalignment.

After applying the above gating steps, a filtered set of valid objects is obtained for each state. Let $\mathcal{O}_k$ represent the set of valid objects associated with state k. Only objects in $\mathcal{O}_k$ are used to construct the residuals in the

置信度评分。仅将评分超过预设阈值的对象纳入后续处理流程。该步骤可剔除低置信度检测结果，这些结果可能为假阳性或定位不准确。

随后通过轨迹长度评估轨迹稳定性。轨迹极短的物体将被排除，因其通常对应于新初始化或不稳定轨迹。通过强制设定最小轨迹长度要求，系统更倾向于选择具有足够时间一致性的物体。

此外，系统引入运动一致性约束以过滤异常目标。对于速度变化幅度过大的物体将被剔除，此类情况通常表明存在跟踪失败、身份切换或运动估计错误等问题。通过剔除这些异常值，剩余的物体观测数据将更可靠地用于轨迹优化。

系统同时考虑自我状态与被追踪对象的时间对齐性。在滑动窗口的每个状态周期内，系统会在限定时间窗口内搜索最接近的被追踪对象信息。若在允许的时间差范围内未找到匹配结果，则相应对象观测数据将被舍弃。该机制可有效避免因时间对齐偏差导致的空间不一致性问题。

在应用上述门控步骤后，可为每个状态获得一组经过过滤的有效对象。设$\mathcal{O}_k$表示与状态k相关联的有效对象集合。仅使用$\mathcal{O}_k$中的对象来构建残差项。

object observation consistency term.

The main gating parameters used in the implementation are summarized in Table 5-2.

Table 5-2 Target gating parameters used in the back-end module

Parameter	Description	Value
$s_{\min}$	Minimum object confidence score	0.6
$\ell_{\min}$	Minimum track length	5
$\Delta t_{\max}$	Maximum allowed time difference	0.1s
$\Delta v_{\max}$	Maximum allowed velocity change	2.0m/s

Based on the gated object set, the back-end optimization proceeds as follows. First, a sliding window is constructed from the buffered odometry states. For each state in the window, temporally aligned tracked objects are retrieved and filtered according to the gating criteria described above. Then, odometry consistency residuals and object observation residuals are constructed for all valid states and objects.

After the residuals are assembled, nonlinear least-squares optimization is performed to update the ego states in the window. The optimization is carried out using a robust loss function to suppress the influence of remaining outliers. Once the optimization converges, the refined ego trajectory is obtained.

The optimized trajectory is then published for evaluation and analysis. It should be noted that this refined trajectory is not fed back into the front-end mapping process, but is used as an improved estimation for performance

客观观察一致性术语

本实施方案中使用的主要门控参数总结如下

表5-2

表5-2 后端模块中使用的靶向门控参数

参数	描述	值
$s_{分}$	最低对象置信度评分	0.6
$\ell_{\min}$	最小轨道长度	5
$\Delta t_{\max}$	最大允许时间差	0.1s
$\Delta v_{\max}$	最大允许速度变化	2.0m/s

基于门控对象集，后端优化流程如下：首先，从缓冲的里程计状态中构建滑动窗口。针对窗口内的每个状态，提取时间对齐的跟踪对象，并根据上述门控标准进行过滤。随后，为所有有效状态和对象构建里程计一致性残差与对象观测残差。

残差数据完成组装后，系统采用非线性最小二乘优化算法更新窗口内的自我状态。该优化过程通过鲁棒损失函数实现，有效抑制剩余离群值的影响。当优化收敛后，即可获得精炼后的自我轨迹。

随后将优化后的轨迹数据对外发布以供评估与分析。需注意的是，该精炼轨迹数据不会反馈至前端映射流程，而是作为性能改进的估算依据使用。

assessment in dynamic environments.

Through this gating and optimization procedure, unreliable object observations are effectively filtered out, and only stable dynamic-scene information is used to assist ego trajectory refinement.

动态环境中的评估

通过这种门控与优化流程，不可靠的物体观测数据被有效过滤，仅利用稳定的动态场景信息来辅助自我轨迹优化。

Chapter 6 Experiment and Analysis

Section 6.1 Introduction to Experimental Data

Our method was evaluated using the KITTI dataset. The KITTI dataset (Karlsruhe Institute of Technology and Toyota Technological Institute) [17] represents one of the most benchmark multimodal sensor datasets in autonomous driving research, jointly released by the Karlsruhe Institute of Technology (Germany) and the Toyota Technological Institute at Chicago. It has been widely adopted in autonomous driving and computer vision studies. The dataset encompasses extensive driving scenarios captured in urban environments, including city streets, residential areas, and highways.

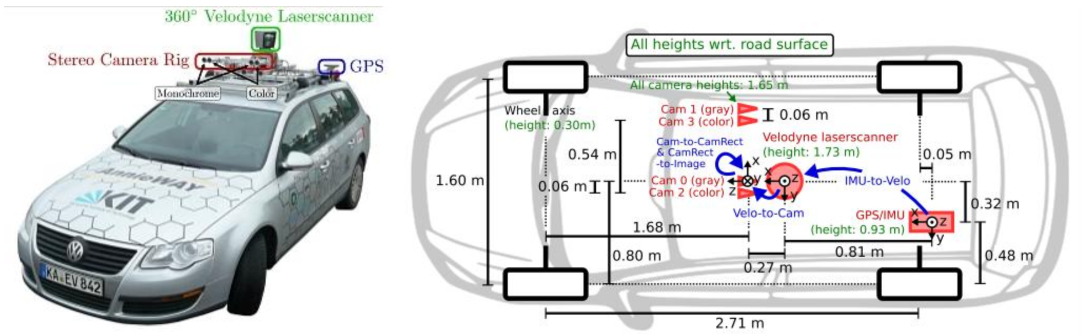

Figure 6-1 The KITTI data collection vehicle

The KITTI dataset provides multiple sensor modalities, including LiDAR point clouds, stereo images, GPS positioning information, and IMU measurements. In the present work, the proposed framework mainly uses LiDAR and IMU data for trajectory estimation and dynamic-scene optimization. The LiDAR point clouds provide the three-dimensional geometric observations

第六章 实验与分析

第6.1节 实验数据概述

我们的方法通过 KITTI 数据集进行了评估。该 KITTI 数据集（卡尔斯鲁厄理工学院与丰田技术研究所联合发布）[17]是自动驾驶研究领域最具代表性的多模态传感器数据集之一，由德国卡尔斯鲁厄理工学院与美国芝加哥丰田技术研究所共同推出。该数据集已被自动驾驶与计算机视觉研究领域广泛采用，涵盖城市街道、居民区及高速公路等多样化驾驶场景。

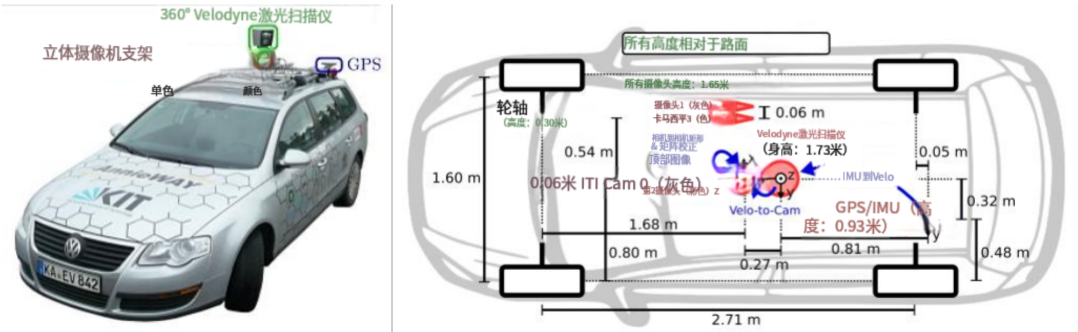

图6-1 KITTI 数据采集车辆

KITTI 数据集包含多种传感器模态数据，包括激光雷达点云、立体图像、GPS定位信息及惯性测量单元（IMU）测量数据。本研究提出的框架主要利用激光雷达与IMU数据进行轨迹估计和动态场景优化。激光雷达点云可提供三维几何观测数据。

of the surrounding environment, which are used for front-end registration in FAST-LIO2 as well as object detection and tracking in the dynamic-scene pipeline. The IMU data supplies high-frequency motion information for state prediction and motion compensation. In addition, the ground-truth trajectory provided by the dataset is used for quantitative evaluation of the estimated results. Through these data sources, the KITTI dataset offers an effective benchmark for testing the proposed method under realistic autonomous driving conditions.

Section 6.2 Experimental Evaluation on KITTI Dataset

Section 6.2.1 Error Analysis Before and After Optimization

To evaluate the effectiveness of the proposed method, comparative experiments were conducted on KITTI tracking sequences. Here, the baseline denotes the trajectory estimated by the system without dynamic weighting and joint back-end optimization, while the optimized result denotes the trajectory estimated with the proposed optimization method enabled.

这些数据来源于周围环境，既用于FAST-LIO2系统的前端注册，也用于动态场景处理流程中的目标检测与跟踪。惯性测量单元（IMU）数据为状态预测和运动补偿提供高频运动信息。此外，数据集提供的真实轨迹数据用于对估计结果进行定量评估。通过这些数据源， KITTI 数据集为在真实自动驾驶条件下测试所提方法提供了有效基准。

第6.2节 KITTI 数据集的实验评估

第6.2.1节 优化前后的误差分析

为评估所提方法的有效性，我们在 KITTI 跟踪序列上进行了对比实验。其中，基线轨迹指系统在未采用动态加权和联合后端优化时估计的轨迹，而优化结果则指启用所提优化方法后估计的轨迹。

Table 6-1 Quantitative comparison between baseline and optimized results

Sequence		ATE RMSE (m)	RPE RMSE (m)	Translation Error (%)	Rotation Error (deg/m)
0003	Baseline	0.5294	0.2049	1.4794	1.1509
	Optimized	0.4513	0.1809	1.3516	1.1569
0006	Baseline	0.5238	0.1324	4.1614	12.8416
	Optimized	0.4710	0.0869	4.0605	11.6291
0009	Baseline	1.6831	0.1787	2.5630	3.2425
	Optimized	1.6675	0.1651	2.5281	3.2294
0010	Baseline	3.7651	0.2774	1.3884	1.1274
	Optimized	0.8645	0.2278	0.9774	1.1353
0013	Baseline	0.6047	0.1495	2.4374	2.3026
	Optimized	0.5816	0.1392	2.3741	2.2419
0020	Baseline	11.7323	0.2250	3.2245	1.1894
	Optimized	6.2936	0.2246	2.2896	1.1517

As shown in Table 6-1, the optimized results achieve lower ATE RMSE than the baseline on all tested sequences listed in this section. Averaged over these sequences, the ATE RMSE is reduced by 25.50%, while the translation error is reduced by 12.27%. The RPE RMSE and rotation error are also reduced by 13.11% and 2.41%, respectively. These results indicate that the proposed method improves trajectory estimation accuracy in both absolute and relative evaluation.

Among the tested sequences, the improvement is most obvious on Sequences 0010 and 0020. For Sequence 0010, the ATE RMSE decreases from 3.7651 m to 0.8645 m. For Sequence 0020, it is reduced from 11.7323 m to

表6-1 基线值与优化结果的定量比较

序列		自动试验设备 RMSE (m)	RPE RMSE（米）	平移误差（%）	旋转误差（度/米）
0003	基线	0.5294	0.2049	1.4794	1.1509
	优化的	0.4513	0.1809	1.3516	1.1569
0006	基准	0.5238	0.1324	4.1614	12.8416
	优化的	0.4710	0.0869	4.0605	11.6291
0009	基线	1.6831	0.1787	2.5630	3.2425
	优化的	1.6675	0.1651	2.5281	3.2294
0010	基线	3.7651	0.2774	1.3884	1.1274
	优化的	0.8645	0.2278	0.9774	1.1353
0013	基线	0.6047	0.1495	2.4374	2.3026
	优化的	0.5816	0.1392	2.3741	2.2419
0020	基线	11.7323	0.2250	3.2245	1.1894
	优化的	6.2936	0.2246	2.2896	1.1517

如表6-1所示，优化结果在本节列出的所有测试序列上均实现了比基线更低的平均绝对误差（ATE）RMSE 。这些序列的平均ATE RMSE 降低了25.50%，翻译误差减少了12.27%。相对位置误差（RPE）RMSE 和旋转误差分别降低了13.11%和2.41%。这些结果表明，所提出的方法在绝对和相对评估指标上均提升了轨迹估计精度。

在测试序列中，改进效果在序列0010和0020上最为显著。序列0010的平均绝对误差（ATE）RMSE 从3.7651米降至0.8645米；序列0020的平均绝对误差则从11.7323米降低至

6.2936 m. These results show that the proposed method can effectively reduce trajectory deviation in some dynamic driving scenes.

For Sequences 0003, 0006, 0009, and 0013, the improvements are relatively smaller, but the optimized results still remain better than the baseline in terms of ATE RMSE. A possible reason is that these sequences contain less dynamic interference, so the influence of the proposed optimization is more limited. Even so, the results still indicate that the method can provide stable improvement in several representative cases.

Overall, the results in Table 6-1 demonstrate that the proposed method improves trajectory estimation accuracy on the selected KITTI tracking sequences. The improvement is more evident in sequences with stronger dynamic interference, while in less dynamic scenes the gain is relatively moderate.

Section 6.2.2 Case Analysis and Study

To further examine the effect of the proposed method in a representative dynamic scenario, Sequence 0020 is selected for case analysis. This sequence was also used as an important reference during parameter tuning. Compared with many other sequences, Sequence 0020 contains stronger dynamic interference, especially a large number of surrounding vehicles with relatively high speed. Therefore, it provides a suitable test case for evaluating the

6.2936米。这些结果表明，所提出的方法能够在某些动态驾驶场景中有效降低轨迹偏差。

对于序列0003、0006、0009和0013，改进幅度相对较小，但在ATE RMSE方面优化结果仍优于基线水平。可能的原因是这些序列包含较少的动态干扰，因此所提优化方案的影响更为有限。即便如此，结果仍表明该方法能在多个代表性案例中实现稳定改进。

总体而言，表6-1的结果表明，所提出的方法提高了所选KITTI跟踪序列中的轨迹估计精度。在动态干扰较强的序列中，改进效果更为显著；而在动态干扰较弱的场景中，增益提升幅度相对有限。

第6.2.2节 病例分析与研究

为深入探究所提方法在典型动态场景中的效果，本研究选取序列0020进行案例分析。该序列在参数调优过程中亦作为重要参考依据。与其他多数序列相比，序列0020具有更强的动态干扰特征，尤其表现为大量高速行驶的周边车辆。因此，该序列可作为评估方法性能的适宜测试案例。

proposed trajectory optimization framework in a challenging traffic environment.

The quantitative results on Sequence 0020 show a clear improvement after optimization. As listed in Table 6-1, the ATE RMSE decreases from 11.7323 m to 6.2936 m after optimization, while the translation error is reduced from 3.2245% to 2.2896%. The RPE RMSE also shows a slight reduction, from 0.2250 m to 0.2246 m. These results indicate that the optimized method can reduce the overall trajectory deviation in this highly dynamic sequence.

在复杂交通环境中提出的轨迹优化框架

序列0020的定量分析结果表明优化后性能显著提升。如表6-1所示，优化后平均轨迹误差（ATE）RMSE 从11.7323米降至6.2936米，平移误差从3.2245%下降至2.2896%。相对位置误差（RPE）RMSE 也略有降低，从0.2250米降至0.2246米。这些结果表明，优化方法可有效降低该高动态序列中的整体轨迹偏差。

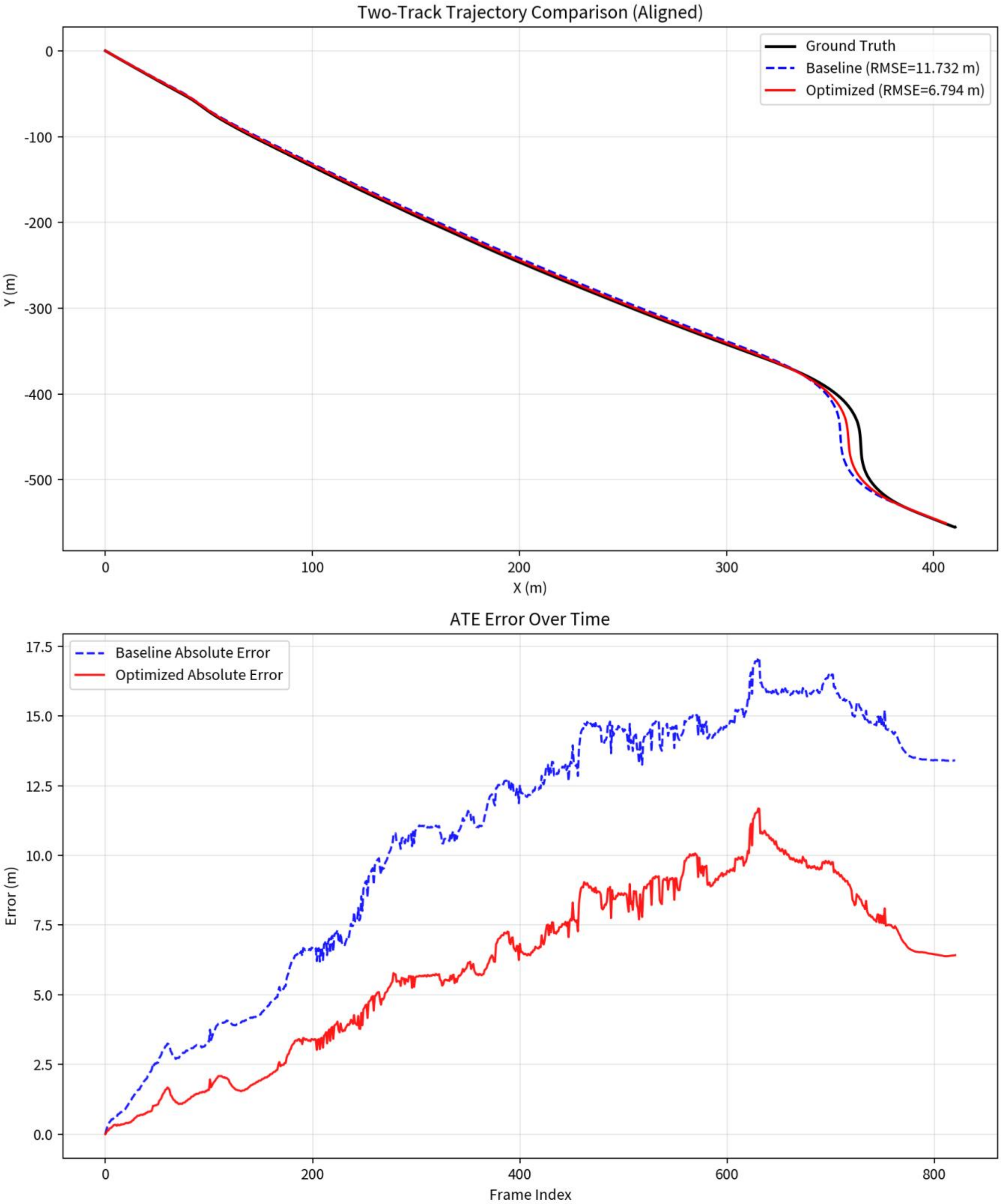

Figure 6-2 Trajectory comparison and ATE error on Sequence 0020

The trajectory comparison in Figure 6-2 further illustrates this improvement. In the aligned trajectory plot, the optimized trajectory remains closer to the ground-truth curve than the baseline, especially in the later part of the sequence

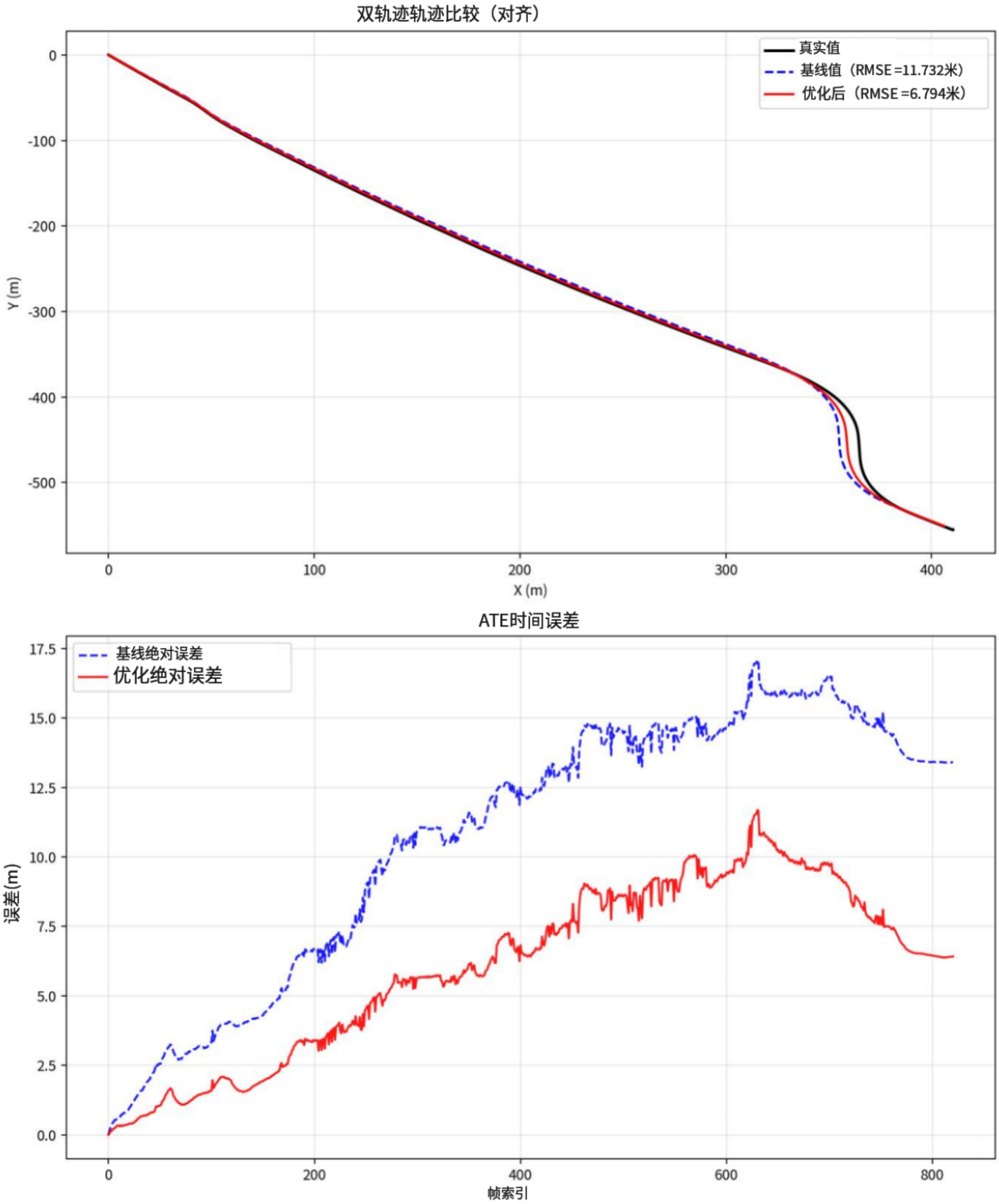

图6-2 序列0020的轨迹比较及ATE误差

图6-2中的轨迹比较进一步证实了这一改进效果。在对齐轨迹图中，优化后的轨迹相较于基线轨迹始终更接近真实曲线，尤其在序列后期阶段表现更为显著。

where the road geometry changes more obviously. The deviation of the baseline trajectory becomes more apparent near the turning region, while the optimized result follows the ground truth more closely. This suggests that the proposed method is able to better suppress trajectory drift under dynamic interference.

The lower plot in Figure 6-2 presents the ATE error over time. It can be observed that the baseline error increases continuously as the sequence proceeds, and reaches a relatively high level in the later frames. By contrast, the optimized result maintains a consistently lower error throughout most of the sequence. Although the optimized trajectory still shows some fluctuation, its error accumulation is clearly slower than that of the baseline. This trend is consistent with the reduction in ATE RMSE reported in Table 6-1.

Figure 6-3 Visualization of tracked dynamic vehicles on Sequence 0020

Figure 6-3 provides additional visualization of the dynamic scene in Sequence 0020. As shown in the point cloud and tracking results, multiple

道路几何形态变化更为显著。基线轨迹的偏差在转弯区域附近表现得更为明显，而优化结果与真实轨迹的吻合度更高。这表明所提出的方法能够在动态干扰环境下更有效地抑制轨迹漂移。

图6-2下半部分展示了随时间变化的平均绝对误差（ATE）变化趋势。可以看出，基准误差随着序列推进持续上升，在后期帧中达到较高水平。相比之下，优化结果在整个序列中大部分时间都保持较低误差。尽管优化轨迹仍存在波动，但其误差累积速度明显慢于基准值。这一趋势与表6-1中报告的ATE RMSE 下降趋势一致。

图6-3 序列0020中追踪动态车辆的可视化呈现

图6-3对序列0020中的动态场景进行了进一步可视化展示。如点云及跟踪结果所示，存在多个

vehicles are distributed around the ego vehicle, and several of them appear in nearby lanes with clear relative motion. Such a scene introduces strong dynamic disturbance to LiDAR-based trajectory estimation. In this case, the proposed framework benefits from the use of tracked dynamic-object information, which helps reduce the negative influence of moving vehicles on trajectory estimation.

Overall, the case study on Sequence 0020 demonstrates that the proposed method is particularly effective in dynamic highway-like traffic scenes with many fast-moving vehicles. Both the quantitative metrics and the visualization results show that the optimized method can produce a trajectory closer to the ground truth and reduce error accumulation over time. This case further supports the effectiveness of the proposed framework for dynamic-scene trajectory estimation.

车辆在目标车辆周围分布，其中多辆车辆以明显相对运动出现在邻近车道。此类场景会对基于激光雷达的轨迹估计产生显著动态干扰。本研究提出的框架通过利用动态目标跟踪信息，有效降低了移动车辆对轨迹估计的负面影响。

总体而言，序列0020的案例研究表明，所提出的方法在包含大量快速移动车辆的动态高速公路类交通场景中表现尤为突出。定量指标与可视化结果均表明，优化方法能生成更接近真实轨迹的运动轨迹，并有效抑制误差随时间累积。该案例进一步验证了所提框架在动态场景轨迹估计中的有效性。

Chapter 7 Summary and Future Plans

This thesis studies trajectory optimization for FAST-LIO2 in dynamic scenes. Focusing on the influence of moving vehicles on LiDAR-inertial odometry, this work builds a dynamic-scene trajectory optimization framework by integrating 3D object detection, multi-object tracking, front-end dynamic weighting, and back-end trajectory refinement. Within this framework, PointPillars is used to provide 3D object detections, MCTrack is used to maintain temporal consistency of surrounding vehicles, and the tracked-object information is further incorporated into the FAST-LIO2 pipeline for dynamic-scene-aware trajectory estimation.

On the front end, tracked objects are used to identify dynamic regions and assign adaptive weights to LiDAR points, so that the negative influence of moving objects on odometry estimation can be reduced without directly removing dynamic observations. On the back end, a joint optimization module is introduced to refine the ego trajectory by combining odometry consistency and object observation consistency within a sliding window. In addition, a semi-synchronous offline frame scheduling mechanism is designed to improve the stability of the full experimental pipeline. Through these designs, the proposed method extends the original FAST-LIO2 framework to a more structured dynamic-scene trajectory optimization system.

Experiments on KITTI tracking sequences show that the proposed method

第七章 总结与未来规划

本论文研究了动态场景中FAST-LIO2系统的轨迹优化问题。针对移动车辆对激光雷达惯性里程计的影响，我们通过整合三维目标检测、多目标跟踪、前端动态加权及后端轨迹优化等技术，构建了动态场景轨迹优化框架。该框架采用PointPillars实现三维目标检测，MCTrack确保周边车辆的时间一致性，同时将目标跟踪信息整合至FAST-LIO2流程中，从而实现动态场景感知的轨迹估计。

在前端处理阶段，通过使用轨迹跟踪对象来识别动态区域，并为激光雷达点分配自适应权重，从而在不直接剔除动态观测数据的情况下，有效降低移动物体对里程计估计的负面影响。后端处理方面，引入联合优化模块通过滑动窗口内结合里程计一致性与物体观测一致性来优化本体轨迹。此外，还设计了半同步离线帧调度机制以提升完整实验流程的稳定性。通过这些设计，本文提出的方法将原始FAST-LIO2框架扩展为更具结构化的动态场景轨迹优化系统。

基于 KITTI 跟踪序列的实验表明，所提出的方法

can improve trajectory estimation accuracy on several representative sequences. In particular, the improvement is more evident in sequences with stronger dynamic interference, as demonstrated by the quantitative comparison and the case study on Sequence 0020. These results indicate that the proposed framework is effective for reducing trajectory deviation in dynamic driving environments.

Although the current framework achieves encouraging results on some representative sequences, there is still considerable room for further improvement. First, the current PointPillars-based detection module mainly focuses on the forward field of view, which limits the completeness of dynamic-object perception in the full surrounding scene. In future work, the detection range and training strategy can be further improved by modifying the detection setting and retraining the detection network, so that more complete target information can be provided for downstream tracking and trajectory optimization.

Second, the current joint back-end module still has limitations. At present, it mainly refines the ego trajectory, while the tracking results of MCTrack themselves are not further optimized in the back-end process. In addition, the optimized trajectory is currently used only for trajectory evaluation and is not fed back to FAST-LIO2 for map update and front-end refinement. In future work, a tighter coupling strategy can be explored, so that the back-end

该方法可在多个代表性序列中提升轨迹估计精度。定量对比分析及序列0020案例研究显示，动态干扰更强的序列中改进效果尤为显著。结果表明，所提出的框架能有效降低动态驾驶环境中的轨迹偏差。

尽管当前框架在部分代表性序列上取得了令人鼓舞的成果，但仍存在显著改进空间。首先，基于PointPillars的检测模块目前主要聚焦于正向视野范围，这限制了对完整周边场景中动态物体感知的全面性。未来研究可通过优化检测参数设置并重新训练检测网络，进一步提升检测范围和训练策略，从而为下游跟踪与轨迹优化提供更完整的目标信息。

其次，当前联合后端模块仍存在局限性。目前该模块主要负责自车轨迹优化，而MCTrack自身的跟踪结果在后端处理过程中未进行进一步优化。此外，优化后的轨迹目前仅用于轨迹评估，未反馈至FAST-LIO2系统进行地图更新及前端优化。未来研究可探索更紧密的耦合策略，以提升后端模块的协同性能。

optimization result can be further integrated with both target tracking and mapping.

Third, the current experimental evaluation and parameter tuning remain limited. Since the performance of the framework still varies across different KITTI sequences, more systematic experiments are needed in future work to analyze the sensitivity of the method to scene dynamics, tracking quality, and parameter settings. A more comprehensive tuning and validation process would help improve the robustness and generalization ability of the proposed framework.

Overall, this work provides an initial exploration of dynamic-scene trajectory optimization for FAST-LIO2. It demonstrates that tracked dynamic-object information can be incorporated into both front-end and back-end processing to improve trajectory estimation in some challenging driving scenarios. Future improvements in detection, back-end coupling, and experimental validation are expected to further enhance the practicality and stability of the framework.

优化结果可进一步与目标跟踪及地图构建相结合。

第三，当前的实验评估与参数调优仍存在局限性。由于该框架在不同KITTI 序列中的表现仍存在差异，未来研究需开展更系统化的实验，以分析该方法对场景动态变化、跟踪质量及参数设置的敏感性。通过更全面的调优与验证流程，可有效提升所提框架的鲁棒性与泛化能力。

总体而言，本研究为FAST-LIO2系统中的动态场景轨迹优化提供了初步探索。研究表明，通过将动态目标跟踪信息整合到前端与后端处理环节，可在某些复杂驾驶场景中显著提升轨迹估计精度。未来若能在目标检测精度、后端系统协同性及实验验证方面取得突破，该框架的实用性和稳定性将得到进一步增强。

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